

Spatial and genetic constraints govern transcription–translation coupling and mRNA degradation in bacteria

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Bacterial gene expression is thought to involve tightly coupled transcription, translation and mRNA degradation. However, recent work has indicated that this is not always the case, leaving the generality and regulation of this coordination unclear. Here we use genetic, kinetic and spatial analyses in *Escherichia coli* to show that transcription–translation coupling requires high translational activity and that nearly half of the transcriptome exhibits signatures consistent with partial uncoupling. We find that co-transcriptional mRNA degradation is rare due to membrane localization of RNase E, except for transcripts encoding inner-membrane proteins. Our results show that translation efficiency determines the level of premature transcription termination, which in turn shapes mRNA degradation patterns and kinetics. Comparative analyses in *Bacillus subtilis* and *Caulobacter crescentus* also reveal species-specific coordination strategies. This challenges the universality of co-transcriptional coupling and defines how spatial and genetic features coordinate bacterial gene expression.

Unlike eukaryotic cells, bacterial cells lack nuclei, and the transfer of genetic information from DNA to protein takes place within a shared space, the cytoplasm. This arrangement permits the concurrent transcription, translation and degradation of an mRNA, thus creating a potential for dynamic gene regulation through direct coupling of these processes. Decades of research in a model bacterium, *Escherichia coli*, has established a paradigmatic notion that an RNA polymerase (RNAP) is physically or functionally coupled to the pioneering ribosome during transcription^{1–3}, that mRNA degradation can start before transcription is completed^{4,5}, and that translating ribosomes protects mRNA from degradation⁶. However, recent studies have challenged the broad applicability of transcription–translation coupling across all genes in *E. coli* as well as across other bacterial species, calling for a re-evaluation of the overall paradigm of bacterial gene expression.

For example, in vivo studies have shown that transcription and translation are not obligatorily coupled at the functional level during elongation in *E. coli*⁷ and that RNAP can outpace ribosomes in *Bacillus subtilis*⁸, indicating that bacterial RNAPs are not continuously accompanied by a leading ribosome. It remains unknown whether transcription–translation coupling varies with any gene-specific or species-specific features.

Co-transcriptional mRNA degradation has also been proposed in *E. coli*^{4,5}, yet its feasibility remains in question because the RNA degradosome—which contains RNase E, the main ribonuclease controlling mRNA degradation^{9,10}—is anchored to the inner membrane of the cell, away from the nucleoid^{11,12}. If co-transcriptional degradation occurs, it could strongly influence how many proteins are produced per transcript and how quickly protein synthesis stops after transcriptional

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repression in response to changing cellular needs. Beyond *E. coli*, RNase E in *Caulobacter crescentus* and other α -proteobacteria is localized in the cytoplasm^{13,14}, raising the possibility that coordination between transcription and mRNA degradation differs across species.

Translation has long been proposed to stabilize mRNAs via ribosome shielding^{15–19}, but the responsible feature of translation remains unclear. For example, it is not known whether mRNA protection depends primarily on ribosome loading near the 5' end or on ribosome occupancy along the transcript. It is also unclear whether this mechanism explains genome-wide variation in mRNA lifetimes⁵. Elucidating how transcription, translation and mRNA degradation are coordinated during the life cycle of an mRNA sets a fundamental framework for understanding gene regulation in response to different environments and for designing synthetic gene expression systems, where gene output must be precisely controlled.

Here we show that co-transcriptional coupling is not universally maintained across genes and species. In *E. coli*, transcription–translation coupling is not obligatory, but instead is modulated by translation initiation rate, which can depend on ribosome binding site (RBS) sequence. We further find that co-transcriptional mRNA degradation is influenced by RNase E localization and whether the encoded protein undergoes membrane insertion during synthesis. In addition, translation can affect mRNA decay patterns and kinetics through premature transcription termination. Analyses in *B. subtilis* and *C. crescentus* further reveal distinct species-specific modes of coordination among transcription, translation and mRNA degradation. Together, these findings suggest that this coordination depends strongly on spatial organization and translation initiation, both of which vary across species and genetic contexts.

Results

Transcription–translation coupling depends on translation

We first investigated transcription–translation coordination in *E. coli*. We hypothesized that translation initiation rate is important for transcription–translation coupling. If translation initiation is slow and early ribosome progression is delayed, transcription may proceed without a coupled ribosome. To test this, we used a monocistronic *lacZ* gene under the *lac* promoter and constructed two RBS mutants, predicted to reduce translation to 17% and 10% of the original RBS according to an RBS calculator²⁰ (Fig. 1a). After induction with the membrane-permeable inducer, isopropyl β -D-1-thiogalactopyranoside (IPTG), at time zero, 3' *lacZ* mRNA (Z3) and LacZ protein were quantified over time by quantitative PCR with reverse transcription (RT–qPCR) and the β -galactosidase assay (Miller assay), respectively. Despite reduced expression in the weak RBS strains, transcription and translation times were comparable (~150 s) in all *lacZ* constructs (Fig. 1b–d), consistent with kinetic coupling of RNAP–ribosome observed previously^{3,8}. However, this result does not exclude partial uncoupling, since the measured transcription and translation times reflect only the fastest RNAP and ribosomes finishing elongation within the ensemble population.

To determine the percentage of RNAPs that are uncoupled from ribosomes, we examined the percentage of premature transcription termination (PT). Previous studies have shown that transcription–translation uncoupling and subsequent premature transcription termination can result from sudden disruptions in translation, such as nonsense mutations²¹, translation-inhibiting antibiotics^{22,23} and amino acid starvation²⁴. In contrast, RBS strength is a tunable and evolvable feature of gene regulation with genome-wide variability²⁵, whose impact on coupling remains untested.

We quantified PT by measuring 5' and 3' *lacZ* mRNA expression after induction (denoted as Z5 and Z3; Fig. 1a). In the native RBS strain, Z5 and Z3 reached similar steady-state levels (Fig. 1e), indicating negligible PT. However, the weak RBS strain showed ~50% lower Z3 than Z5 (Fig. 1f), suggesting significant PT between the Z5 and Z3 regions. This effect was Rho dependent as treatment with bicyclomycin (BCM),

an inhibitor of the Rho factor²⁶, equalized the steady-state Z5 and Z3 levels (Fig. 1g and Extended Data Fig. 1).

Extending this analysis to additional RBS variants, we constructed eight additional *lacZ* RBS mutants spanning a wide range of RBS strengths, inserting them at either the native locus (inducible by IPTG) or the *araB* locus (inducible by arabinose; sequences in Extended Data Table 1 and protein levels and translation initiation rates in Extended Data Fig. 2a,b). PT showed an excellent negative correlation with the predicted RBS strength ($r = -0.93$; Fig. 1h), supporting the idea that translation initiation rate is a critical factor governing transcription–translation coupling.

To assess generality, we compared genome-wide TE²⁵, a proxy for translation initiation rate per transcript²⁷, with the transcription PF, calculated from simultaneous 5' and 3' end sequencing (SEnd-seq)²⁸ as the ratio of downstream (≥ 500 nt from the transcription start site (TSS)) to upstream (0–200 nt) coverage²⁹ (Extended Data Fig. 2c). By this definition, PF < 1 reflects depletion of downstream reads relative to upstream reads. As an integrative readout of relative RNA abundance along the gene, PF can be influenced by multiple processes, including transcriptional pausing, termination and RNA degradation. Given the RNA pause profile³⁰ and assuming that RNA decay predominantly initiates from the 5' end in *E. coli*^{31,32}, we can show that a low PF value is mainly associated with increased PT (Supplementary Discussion 1). We thus asked whether low-TE genes tend to exhibit lower PF, as expected if translation initiation rate influences the degree of transcription–translation coupling genome-wide. Among the 1,336 genes analysed (median TE = 0.94 and PF = 0.99), high-TE genes showed PF near 1, while lower TE (<1) was associated with progressively reduced PF (Fig. 1i and Extended Data Fig. 2d). Inhibition of the Rho factor by BCM selectively increased PF in low-PF genes (Fig. 1j and Extended Data Fig. 2e,f), consistent with low PF being associated, at least in part, with Rho-dependent premature termination. We note that the PF increase did not reach full recovery (to 1), and this observation may be explained by Rho-independent mechanisms that can also limit RNAP processivity in the absence of RNAP–ribosome coupling³³. Together, these observations are consistent with translation initiation rate being an important kinetic determinant of gene-specific transcription–translation coupling across the *E. coli* genome.

Co-transcriptional degradation is negligible for *lacZ* mRNA

Next, we investigated coordination between transcription and mRNA degradation. Using transient induction of *lacZ*, we measured co-transcriptional and post-transcriptional mRNA degradation rates (k_{d1} and k_{d2} , respectively; Fig. 2a). To do so, glucose was added at 75 s after induction to re-repress transcription before the first RNAPs finished transcription³⁴ (Fig. 2b). Our earlier study³⁴, along with an older report³⁵, indicates that promoter re-repression occurs rapidly, within ~10 s after glucose addition. Until the first RNAPs finish transcription (T_3), all *lacZ* mRNAs are expected to be nascent (time window i). After the last RNAPs finish transcription (t_3), all *lacZ* mRNAs are released (time window iii). By fitting the decay of Z5 in these two windows with an exponential function, we estimated k_{d1} and k_{d2} . Here, decay refers to the loss of mRNA sequences encompassed by the RT–qPCR amplicon, acknowledging that cleavage events elsewhere in the transcript are not detected as the decay of this region.

If co-transcriptional degradation of *lacZ* mRNA takes place ($k_{d1} > 0$), the Z5 level will decrease in the time window i, as demonstrated by a mathematical model (Extended Data Fig. 3a and Supplementary Discussion 2). However, our data showed constant Z5 levels in the time window i, suggesting negligible co-transcriptional degradation (Fig. 2c). Across biological replicates, k_{d1} was $0.042 \pm 0.060 \text{ min}^{-1}$ and k_{d2} was $0.43 \pm 0.07 \text{ min}^{-1}$. The k_{d2} value is comparable to previously reported *lacZ* mRNA lifetimes^{34,36}, but k_{d1} is much lower than expected. Comparable results were obtained when transcription initiation was blocked with rifampicin³⁷ instead of glucose (Extended Data Fig. 3b,c).

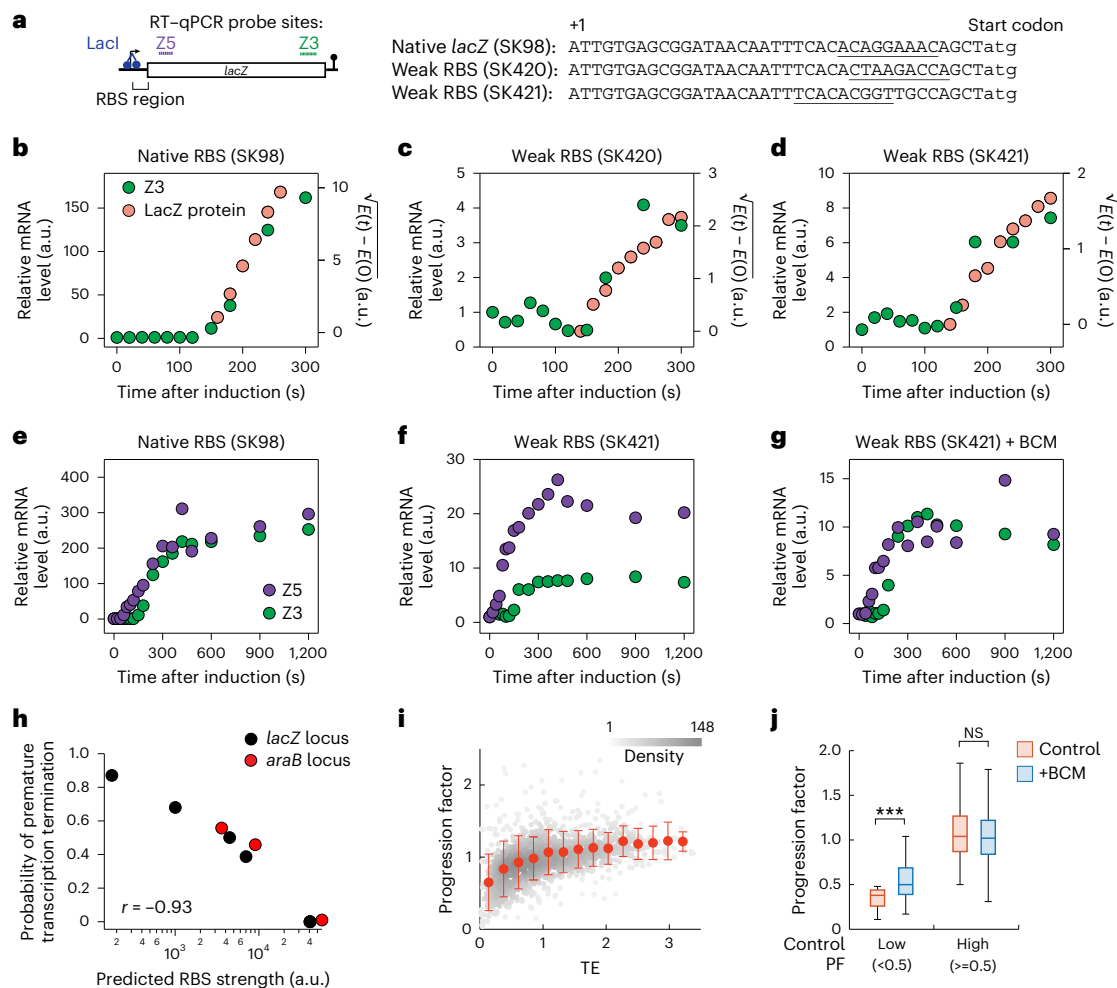


Fig. 1 | Effect of translation activity on transcription-translation coupling.

a, Sequences of *lacZ* mRNA from the first base (+1) to the start codon (atg). Shine-Dalgarno elements estimated by an RBS calculator²⁰ are underlined. **b–d**, Transcription and translation kinetics of *lacZ* after adding 0.2 mM IPTG at time zero for the native RBS SK98 (**b**), weak RBS SK420 (**c**) and weak RBS SK421 (**d**). The 3' *lacZ* mRNA (Z3) was quantified by RT-qPCR using primers amplifying 2,732–2,890 nucleotide (nt) region (gene length, 3,072 nt). LacZ protein was quantified by the Miller assay with background subtraction ($E(t) - E(0)$). Data are shown as mean of two biological replicates. **e–g**, Z5 and Z3 levels following induction with 0.2 mM IPTG at time zero for the native RBS SK98 (**e**), weak RBS SK421 (**f**) and BCM-treated weak RBS SK421 (**g**). In **g**, 100 $\mu\text{g ml}^{-1}$ BCM was added 5 min before IPTG. Z5 was quantified using primers amplifying the 530–660 nt

region; Z3 primers are as in **b–d**. Data are shown as mean of two biological replicates. **h**, Probability of premature transcription termination during *lacZ* expression as a function of RBS strength predicted by an RBS calculator²⁰ (Extended Data Table 1). Data are shown as mean of two biological replicates. **i**, Density scatterplot of translation efficiency (TE) and progression factor (PF) for $n = 1,336$ genes. Red dots represent median $\pm$ s.d. of binned data. A full-scale version is shown in Extended Data Fig. 2d. **j**, PF measured in control (no BCM) and BCM-treated *E. coli* cells, separated by PF in the control sample. The low- and high-PF groups contain 83 and 836 genes, respectively. *** $P = 3.3 \times 10^{-10}$; NS, not significant ($P = 0.092$), from two-sided two-sample t -tests. Boxplots show the median (centre line), interquartile range (box bounds), and minima and maxima (whiskers). *E. coli* cells were grown in M9 glycerol at 30 °C unless otherwise noted.

The estimated nascent mRNA lifetime ($1/k_{d1} \approx 24$ min) greatly exceeded the transcription time (~ 3.5 min), suggesting that *lacZ* mRNA rarely experiences degradation during transcription elongation.

RNase E localization uncouples transcription and mRNA decay

Using a temperature-sensitive RNase E allele³⁸ (*rne3071*), we confirmed that both k_{d1} and k_{d2} of *lacZ* mRNA are controlled by RNase E (Extended Data Fig. 3d). Since RNase E is associated with the inner membrane via its membrane targeting sequence (MTS)^{11,12}, the negligible co-transcriptional degradation of *lacZ* could reflect spatial separation between RNase E and transcription sites in the nucleoid. To test this, we employed a cytoplasmic RNase E mutant (RNase E Δ MTS¹¹) and observed increases in both k_{d1} and k_{d2} ; in particular, k_{d1} increased ~ 7 -fold to $0.31 \pm 0.08 \text{ min}^{-1}$ compared to the wild-type RNase E (Fig. 2d). Another cytoplasmic RNase E mutant, RNase E (1–529), which lacks both the MTS and the RNA degradosome assembly domain¹¹, also increased

k_{d1} and k_{d2} compared to its membrane-bound counterpart, RNase E (1–592) (Extended Data Fig. 3e,f). Altogether, these results show that membrane localization of RNase E limits co-transcriptional mRNA degradation and thereby uncouples transcription from degradation. This principle may apply broadly to many transcripts whose decay is governed by RNase E (see Extended Data Fig. 4 for other example genes).

We next tested whether positioning a nascent transcript near the membrane is sufficient to promote co-transcriptional degradation. Because expression of *lacY* can relocalize its chromosomal region towards the membrane through transertion³⁹, we inserted constitutively expressed *lacY* downstream of *lacZ* (strain SK435; Fig. 2e). As a control, we replaced *lacY* with *aadA*, which encodes a cytoplasmic protein and does not undergo transertion³⁹ (strain SK390). Using fluorescence in situ hybridization (FISH) probes targeting the first 1-kb region of *lacZ* mRNA (Z5_{FISH})⁴⁰, we confirmed that nascent *lacZ* mRNAs were closer to the membrane in SK435 than in SK390 (Extended Data Fig. 5).

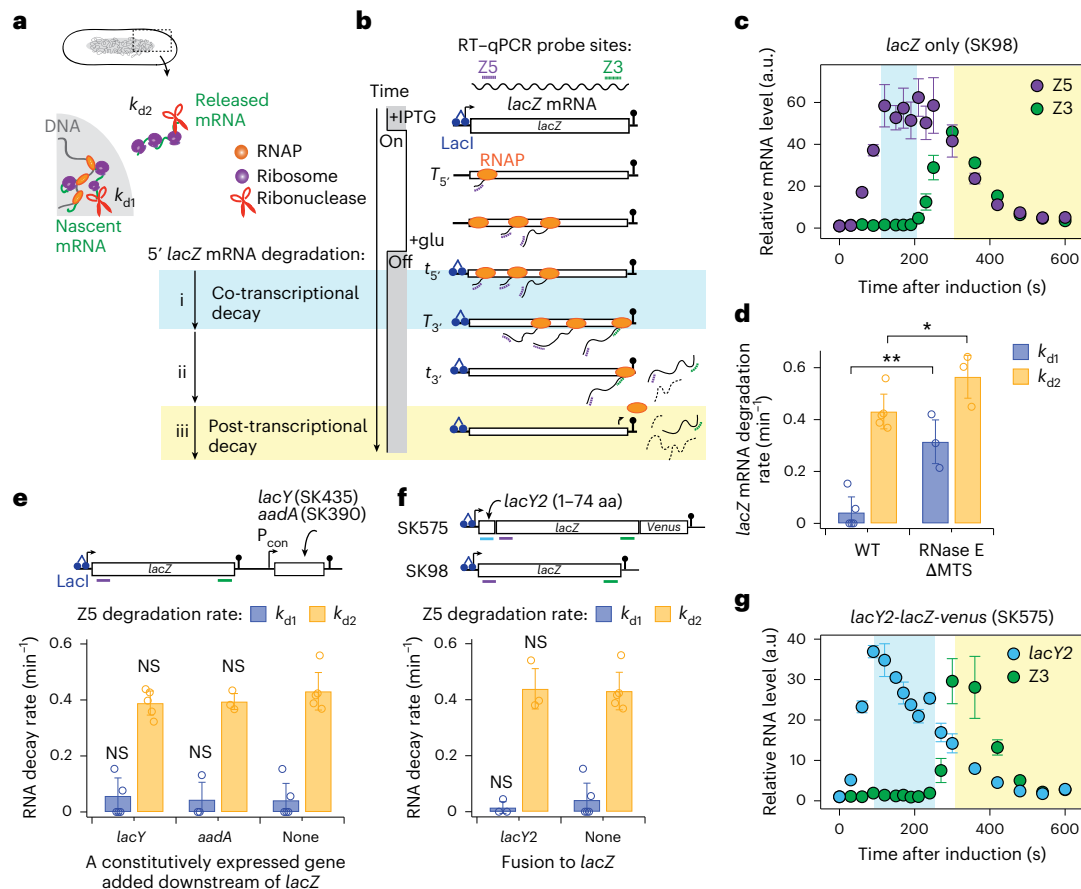


Fig. 2 | Co-transcriptional mRNA degradation in *E. coli*. **a**, Definition of co-transcriptional and post-transcriptional mRNA degradation with rates k_{d1} and k_{d2} , respectively. **b**, Schematic of the time-course assay. *lacZ* transcription was transiently induced, and Z5 and Z3 levels were quantified by RT-qPCR. The first and last RNAPs passed the Z5 probe site at T_5 and T_3 , respectively, and the Z3 probe site at T_3 and T_5 , respectively. **c**, Z5 and Z3 levels following induction with 0.2 mM IPTG at time zero and re-repression with 500 mM glucose at $t = 75$ s. Blue and yellow boxes indicate the time windows i and iii used to determine k_{d1} and k_{d2} of *lacZ* (Z5), respectively. Data are mean $\pm$ s.d. of biological triplicates. **d**, k_{d1} and k_{d2} of *lacZ* mRNA in strains with membrane-bound or cytoplasmic RNase E (SK98 and SK339). Data are mean $\pm$ s.d. of five (SK98) and three (SK339) biological replicates.

replicates. k_{d1} , $**P = 0.0018$; k_{d2} , $*P = 0.037$, from one-sided two-sample *t*-tests.

e, k_{d1} and k_{d2} of *lacZ* mRNA with different downstream genes: *lacY* (SK435), *aadA* (SK390) or none (SK98). Data are mean $\pm$ s.d. of five (SK435, SK98) and three (SK390) biological replicates. k_{d1} , $P = 0.46$ (SK435), $P = 0.49$ (SK390); k_{d2} , $P = 0.84$ (SK435), $P = 0.77$ (SK390), from one-sided two-sample *t*-tests relative to SK98. **f**, k_{d1} and k_{d2} of *lacZ* mRNA with or without translational fusion to *lacY2* (SK575 and SK98). Data are mean $\pm$ s.d. of three (SK575) and five (SK98) biological replicates. k_{d1} , $P = 0.93$; k_{d2} , $P = 0.45$, from one-sided two-sample *t*-tests. **g**, Relative *lacY2-lacZ-venus* (SK575) mRNA level measured by RT-qPCR using primers amplifying the 80–222 nt region. Blue and yellow boxes indicate the time windows i and iii used to determine k_{d1} and k_{d2} of *lacY2*. Data are mean $\pm$ s.d. of biological triplicates.

However, k_{d1} and k_{d2} were similar between strains and almost identical to the original *lacZ*-only strain (Fig. 2e).

To further increase membrane proximity, we constructed a translational fusion of *lacZ* to the first two transmembrane segments of *lacY* (*lacY2*). Venus was fused to the C terminus to visualize the protein product (strain SK575; Extended Data Fig. 6a). The nascent *lacZ* mRNAs showed stronger membrane proximity than in SK435, probably because *lacZ* is fused to *lacY2* as a contiguous transcript (Extended Data Fig. 6b–e), yet k_{d1} for the *lacZ* region remained low and comparable to the original *lacZ*-only strain (Fig. 2f). Thus, membrane proximity of a nascent transcript is not sufficient to promote co-transcriptional decay. One explanation is that membrane-bound RNase E diffuses very slowly¹², limiting its access to nascent RNAs even near the membrane, whereas freely diffusing cytoplasmic RNase E more readily engages nascent RNAs, leading to increased k_{d1} .

lacY transcripts experience co-transcriptional degradation

In the *lacY2-lacZ* fusion, we also measured the degradation kinetics of the *lacY2* region. Strikingly, *lacY2* exhibited fast co-transcriptional degradation with $k_{d1} = 0.34 \pm 0.04 \text{ min}^{-1}$, close to its $k_{d2} = 0.47 \pm 0.08 \text{ min}^{-1}$ (Fig. 2g). This result was not due to truncation of *lacY* because a similar

k_{d1} was observed for full-length *lacY* expressed as a monocistronic transcript from P_{lac} (Extended Data Fig. 6f–h). These findings suggest that coupling between transcription and mRNA degradation can occur for transcripts encoding inner-membrane proteins. Because membrane proximity alone did not increase k_{d1} of *lacZ* (Fig. 2e,f), additional factors linked to translation may promote co-transcriptional degradation of *lacY* mRNA (see Discussion).

Weak RBS increases k_{d1} via premature transcript release

We next investigated how translation affects mRNA degradation. Weak RBS sequences are known to destabilize mRNA^{15,16,19}, decreasing protein production by reducing both translation initiation rate and mRNA lifetime. Remarkably, a weak RBS mutant showed ~15-fold increase in k_{d1} ($0.65 \pm 0.17 \text{ min}^{-1}$) without much change in k_{d2} (Fig. 3a and Extended Data Fig. 7a). This high k_{d1} was largely RNase E dependent, as a temperature-sensitive RNase E allele (*rne3071*) showed much lower k_{d1} at the non-permissive temperature compared with the wild-type RNase E (Extended Data Fig. 7b,c).

Considering the PT observed in weak RBS mutants (Fig. 1h), we hypothesized that this high k_{d1} does not reflect true co-transcriptional degradation of RNAP-tethered nascent RNA, but degradation of

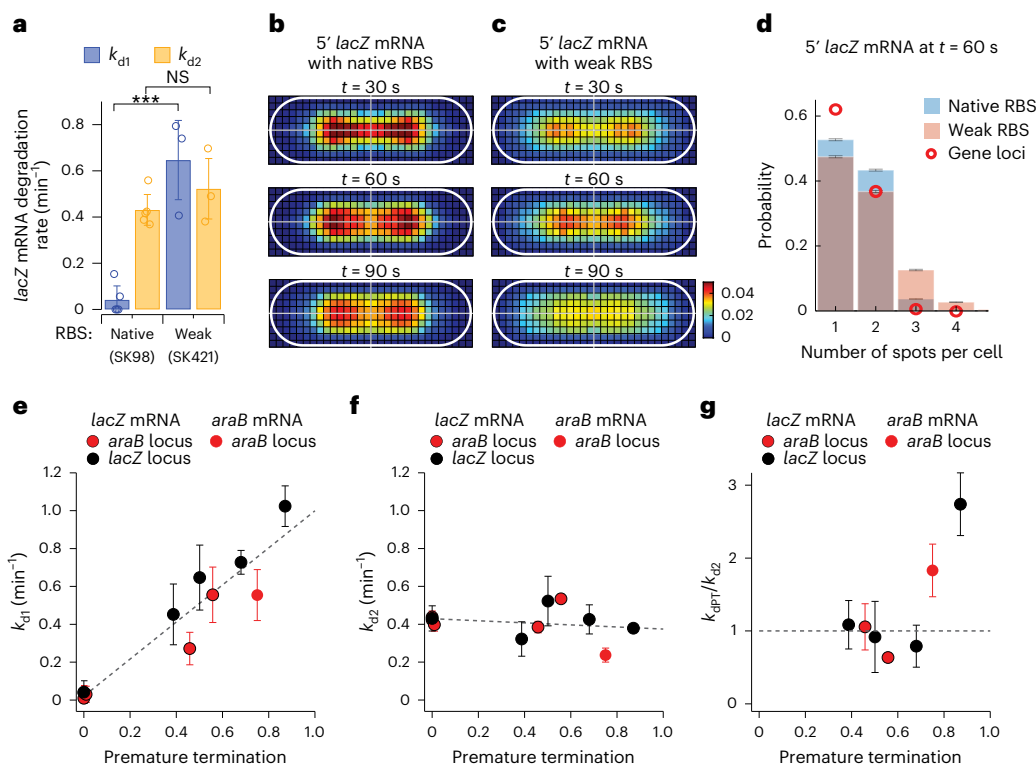


Fig. 3 | Effect of translation on *lacZ* mRNA degradation. **a**, k_{d1} and k_{d2} of *lacZ* mRNA in the native and weak RBS strains. Data are presented as mean $\pm$ s.d. of five (SK98) and three (SK421) biological replicates. k_{d1} , $***P = 3.9 \times 10^{-4}$; k_{d2} , $P = 0.15$, from one-sided two-sample t -tests. **b, c**, 2D histograms of $Z5_{FISH}$ localization from the native (**b**) and weak RBS (**c**) strains (SK519 and SK591). Colour indicates the probability of finding $Z5_{FISH}$ in each bin. Foci positions were normalized in the first quartile and mirrored across cell axes. Bin size, 70–80 nm. White outlines indicate cell boundaries; white lines are axes of symmetry. The same colour scale was applied to all histograms. For the time course, *lacZ* expression was induced with 0.2 mM IPTG at time 0 and re-repressed with 500 mM glucose at $t = 50$ s. At each time point, >25,000 foci were analysed from 5 biological replicates.

d, Number of $Z5_{FISH}$ spots detected per cell at $t = 60$ s in the time-course experiment described in **b** and **c**. The analysis included 26,197 cells for the native RBS (SK519) and 24,363 cells for the weak RBS (SK591) from 5 biological replicates. Data are presented as mean $\pm$ s.e. from bootstrapping. *lacZ* gene loci were imaged in a separate strain (SK259)⁷¹ grown under identical conditions at 43.5 °C, and data are presented as the mean across 3,023 cells from 1 biological replicate. **e–g**, Relationship between k_{d1} , k_{d2} or k_{dPT}/k_{d2} and PT for various RBS–*lacZ* mRNAs and P_{ara} –*araB* mRNA. Dashed lines represent linear fits in **e** and **f**, and a reference line at 1 in **g**. k_d data are presented as mean $\pm$ s.d. of five (SK98) and three (all other strains) biological replicates; PT data are presented as the mean of biological duplicates.

prematurely released transcripts present in the time window **i** due to transcription–translation uncoupling. These incomplete transcripts may diffuse freely in the cytoplasm and be degraded by membrane-localized RNase E.

To test this, we examined *lacZ* mRNA localization during the time window **i** by visualizing 5' *lacZ* mRNAs using FISH ($Z5_{FISH}$). Given the high k_{d1} in the weak RBS strain, we expected low signal. Thus, we performed FISH in *rne3071* strains at the non-permissive temperature to stabilize transcripts. At the elevated temperature, transcription elongation was faster, giving $T_{3'} = 100$ s for the time window **i** when glucose was added at $t = 50$ s (Extended Data Fig. 7b). Two-dimensional (2D) histogram of $Z5_{FISH}$ localizations within this time window showed that $Z5_{FISH}$ foci were clustered at specific sites in the native RBS strain while being broadly dispersed in the weak RBS strain (Fig. 3b,c). In addition, the weak RBS strain showed more $Z5_{FISH}$ foci per cell with lower fluorescence intensity (Fig. 3d and Extended Data Fig. 7d), consistent with a larger fraction of transcripts dissociating from the gene locus early in the time window **i**.

We then tested whether the elevated k_{d1} scales with the abundance of prematurely released transcripts. Across engineered *lacZ* RBS variants and a native weak-RBS gene (*araB*), PT strongly correlated with k_{d1} ($r = 0.94$; Fig. 3e and Extended Data Fig. 7e), supporting the idea that high k_{d1} primarily reflects decay of prematurely released transcripts. Consistent with this view, k_{d1} of the weak RBS *lacZ* mutant was low in BCM-treated cells (Extended Data Fig. 7f). We therefore denote the true co-transcriptional mRNA degradation rate as $k_{d1'}$ to differentiate

it from k_{d1} measured from the time window **i**, which can include decay of prematurely released transcripts.

Minimal RBS effect on k_{d2} suggests limited ribosome protection

Ribosomes are widely believed to protect mRNAs from RNase E⁶, predicting shorter lifetimes for transcripts with weak RBS sequences. To test this model across multiple RBS variants, we analysed the relationship between PT and k_{d2} , the decay rate of 5' mRNA after the last RNAP passes the end of the gene (time window **iii**), which largely reflects decay of full-length transcripts. Across 11 strains with distinct RBS sequences, k_{d2} was independent of PT ($r = -0.22$; Fig. 3f). The lack of correlation was also observed between PT and the degradation rate of 3' mRNA (Extended Data Fig. 7g), implying that ribosome occupancy alone does not explain mRNA stability across these constructs.

Although the degradation rate of prematurely terminated transcripts (k_{dPT}) cannot be measured directly, we developed a mathematical model to infer k_{dPT} from steady-state 5' and 3' mRNA levels (Supplementary Discussion 3). One might expect k_{dPT} to exceed k_{d2} because prematurely released transcripts differ from full-length mRNAs in length, ribosome occupancy and absence of a 3' untranslated region (UTR) or a stem-loop at the intrinsic terminator⁴¹. Surprisingly, k_{dPT} was comparable to k_{d2} in low-to-intermediate PT (<0.7) constructs, independent of the PT value (Fig. 3g and Extended Data Fig. 7h). k_{dPT} was higher than k_{d2} only in an extreme case of very high PT (>0.8).

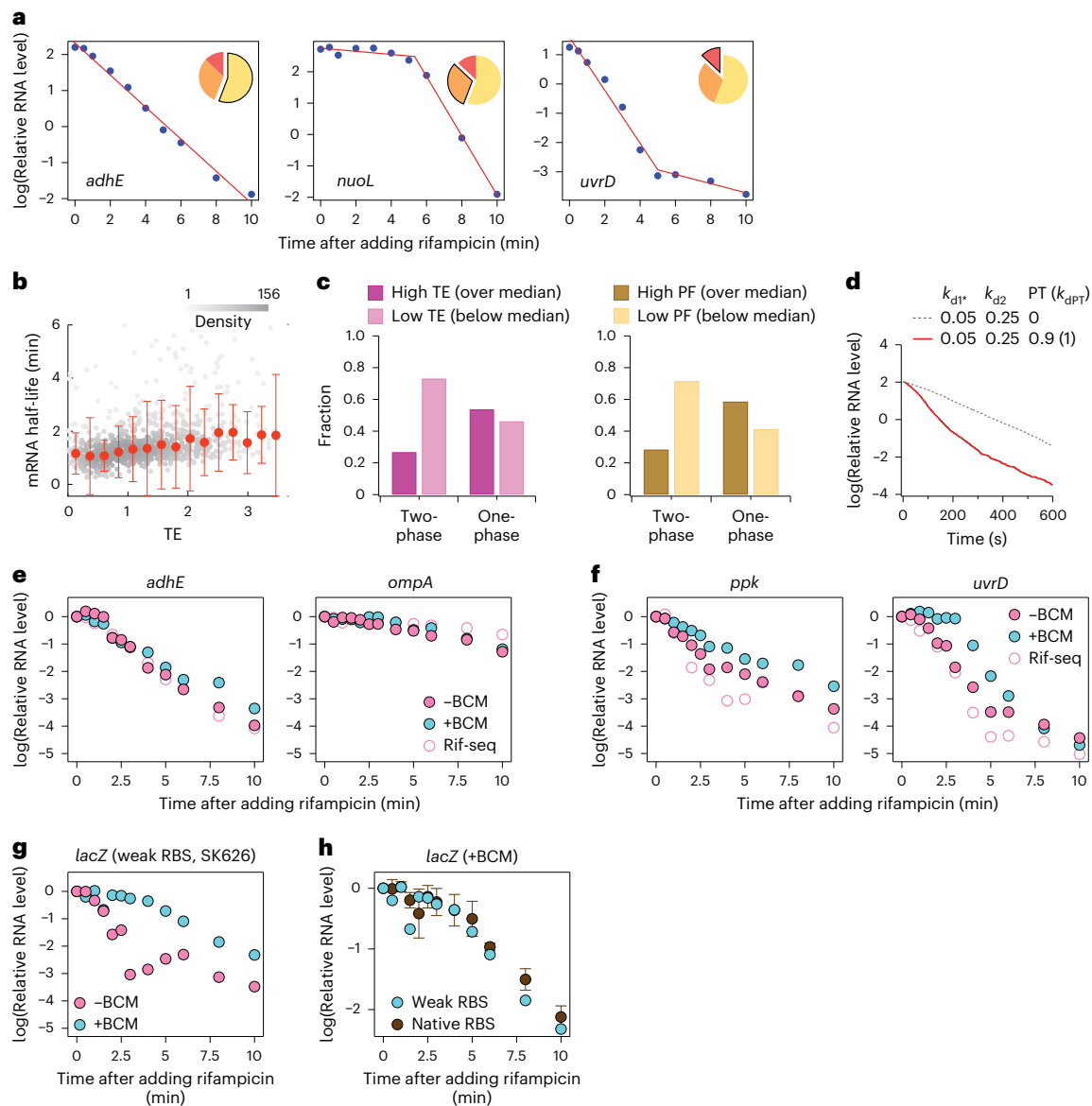


Fig. 4 | Global analysis of the interplay between ribosome occupancy, premature transcription termination and mRNA stability. **a**, Example mRNA decay profiles from Rif-seq. Pie charts show the fraction of genes in each decay-pattern class: one-phase decay (56%; yellow); delayed one-phase decay, defined as a slow phase followed by fast decay (31%; orange); and two-phase decay with an initial fast phase followed by slow decay (13%; red). The black outline marks the class of the example gene shown. Blue circles are Rif-seq data and red lines are piece-wise regression fits. **b**, Density scatterplot of TE and mRNA half-life for one-phase decay genes ($n = 820$). Red dots represent median $\pm$ s.d. of binned data. A full-scale version is shown in Extended Data Fig. 8c. **c**, Non-random association of decay pattern with TE and PF, with two-phase decay genes enriched among low-TE and low-PF genes (Fisher's exact test: $P = 1 \times 10^{-10}$ for TE and 2.4×10^{-6}

for PF). Fractions were calculated from gene counts ($n = 682$ genes total). **d**, Simulated mRNA decay following inhibition of transcription initiation at time zero. The dotted line represents $PT = 0$; the red solid line represents $PT = 0.9$ with $k_{d1} = 4k_{d2}$ (unit: min^{-1}). **e–h**, Decay of selected genes in control (–BCM) and BCM-treated cells measured by RT–qPCR. Rifampicin ($500 \mu\text{g ml}^{-1}$) was added at time 0 to exponentially growing *E. coli* cells cultured in MOPS EZ-rich defined medium at 37°C , matching the conditions used for Rif-seq (SK626 for weak RBS). For *lacZ* expression, 0.2 mM IPTG was added 20 min before rifampicin. When indicated, BCM ($100 \mu\text{g ml}^{-1}$) was added 10 min before rifampicin. RNA levels are normalized to the level at time zero. Data are presented as mean $\pm$ s.d. of biological triplicates for SK98 + BCM and as the mean of biological duplicates for all others. Rif-seq data are shown for comparison.

Thus, except in cases of very weak RBS sequences that may produce ribosome-free mRNAs, degradation rates of full-length and prematurely released transcripts (k_{d2} and k_{dPT}) are largely insensitive to translation initiation rate, refining rather than fully supporting the ribosome shielding model.

Certain low-TE genes show two-phase decay in rifampicin-seq

To investigate the effect of translation on mRNA degradation genome-wide, we performed rifampicin (Rif)-seq, a method that employs rifampicin to stop transcription initiation and measures RNA decay over

time⁵. Unlike the transient induction assay for *lacZ*, Rif-seq measures decay from steady state and cannot directly distinguish k_{d1} , k_{d2} and k_{dPT} .

Using a piece-wise regression model, we found some genes exhibiting a two-phase decay pattern with distinct degradation rates at the early and late stages following rifampicin treatment. Among the 943 genes analysed, 123 exhibited fast-slow two-phase decay, while the remaining 820 showed one-phase decay (Fig. 4a). The one-phase decay group included genes with early slow decay followed by a faster phase, which probably reflects delayed decay onset due to residual transcription elongation because these genes tend to be farther from the TSS

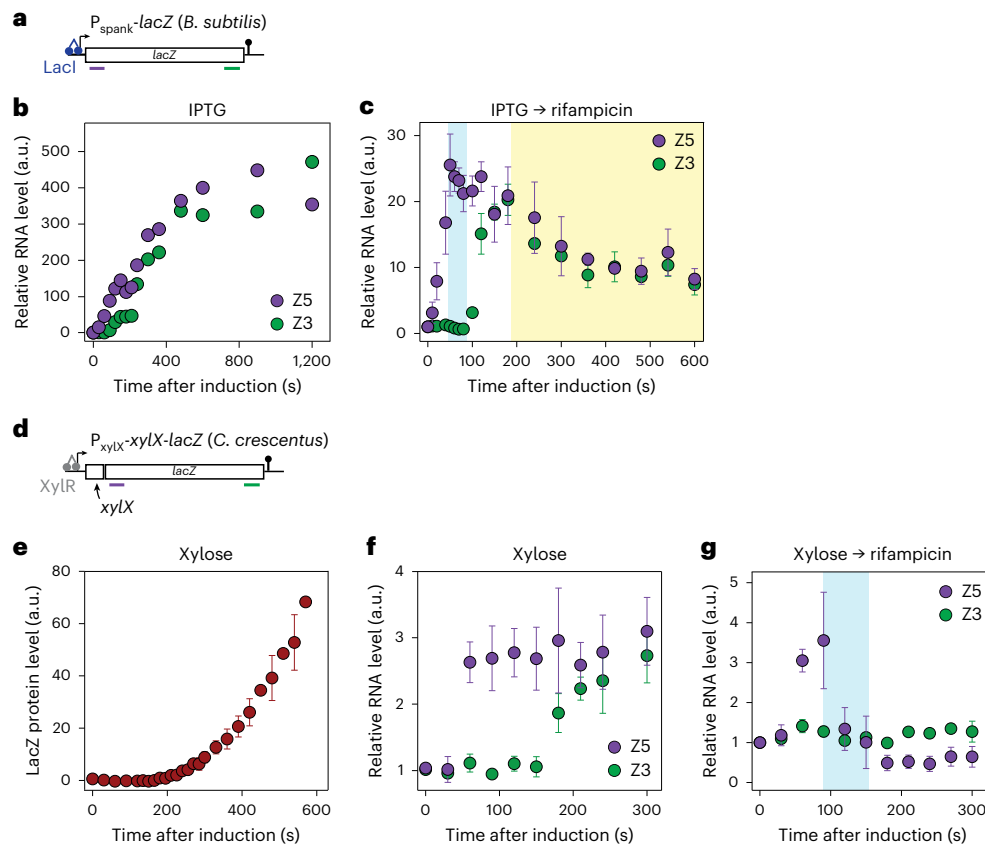


Fig. 5 | Coordination of transcription, translation and mRNA degradation during *lacZ* expression in *B. subtilis* and *C. crescentus*. **a**, IPTG-inducible *lacZ* in the chromosome of *B. subtilis*. For RT-qPCR, the same Z5 and Z3 primers used for *E. coli lacZ* were applied. **b**, **c**, Z5 and Z3 levels after induction with 5 mM IPTG at $t = 0$. To measure *lacZ* mRNA degradation rates in **c**, transcription was re-repressed with rifampicin ($200 \mu\text{g ml}^{-1}$) at $t = 30$ s. Blue and yellow boxes indicate the time windows used for k_{di} and k_{d2} fitting. *B. subtilis* cells were grown at 30 °C in MOPS medium supplemented with maltose. Data in **b** are presented as the mean of biological duplicates; data in **c** are presented as mean $\pm$ s.d. of biological triplicates.

d, Xylose-inducible *lacZ* in *C. crescentus*. For RT-qPCR, the same Z5 and Z3 primers used for *E. coli lacZ* were applied. **e**, Translation kinetics of LacZ protein expression in *C. crescentus* after adding 0.3% xylose, measured by the Miller assay. Data are presented as mean $\pm$ s.d. of biological triplicates. **f**, **g**, Z5 and Z3 levels after induction with 0.3% xylose at $t = 0$. To measure *lacZ* mRNA degradation rates in **g**, transcription was re-repressed with rifampicin ($200 \mu\text{g ml}^{-1}$) at $t = 50$ s. The time window used for k_{di} fitting is indicated by a blue box. *C. crescentus* cells were grown in M2G medium at 28 °C. Data are presented as mean $\pm$ s.d. of seven (**f**) and three (**g**) biological replicates.

(Extended Data Fig. 8a,b). These two-phase decay patterns were also reported in Rif-seq of *Salmonella enterica* as delayed decay onset and slow decay intermediates⁴².

For one-phase decay genes, mRNA half-life showed only a weak positive correlation with TE (Pearson $r = 0.27$, Spearman $\rho = 0.42$; Fig. 4b and Extended Data Fig. 8c), indicating that ribosome occupancy is not a dominant determinant of mRNA stability genome-wide. Likewise, genes within the same operon exhibited similar mRNA decay kinetics despite very different levels of ribosome occupancy (Extended Data Fig. 8d,e). When two-phase decay genes were included, the correlation between TE and RNA half-life was weaker for late-phase half-lives ($r = 0.068$, $\rho = 0.23$; Extended Data Fig. 8f) than for early-phase half-lives ($r = 0.30$, $\rho = 0.46$; Extended Data Fig. 8g).

Strikingly, two-phase decay genes were significantly enriched among low-TE and low-PF genes (Fisher's exact test $P = 1 \times 10^{-10}$ for TE and 2.4×10^{-6} for PF; Fig. 4c), suggesting a link between low translation activity, premature transcription termination and rapid decay of released transcripts. Consistent with this interpretation, simulations predicted that coexistence of RNA isoforms with distinct decay rates (for example, $k_{\text{dPT}} > k_{\text{d2}}$) would generate two-phase decay following rifampicin treatment (Fig. 4d and Extended Data Fig. 8h,i). These genes tended to have short half-lives during the early decay phase, possibly reflecting high k_{dPT} , which may increase the overall TE-mRNA half-life correlation (Extended Data Fig. 8g).

Experimentally, two-phase decay genes with low TE and PF (*ppk* and *uvrD*) exhibited altered decay patterns upon Rho inhibition with BCM, whereas one-phase decay genes with high TE and PF (*adhE* and *ompA*) were unaffected (Fig. 4e,f). A weak-RBS *lacZ* mutant (with $PT = 0.68$, SK626) showed the same BCM-dependent shift in the decay pattern (Fig. 4g). Notably, in BCM-treated cells, decay patterns of the weak- and native-RBS *lacZ* constructs were indistinguishable (Fig. 4h), suggesting that RBS strength has minimal impact on mRNA degradation in the absence of premature transcription termination. Together, these results indicate that rapid decay of prematurely terminated transcripts underlies the two-phase decay pattern and potentially contributes to the modest genome-wide correlation between TE and mRNA half-life.

Coordination of gene expression in other bacteria

Recent studies showed that in *B. subtilis*, RNAP outpaces the ribosome during *lacZ* expression^{8,43}. Whether this uncoupling results in premature termination or elevated k_{di} remains unclear. We measured transcription and translation times of *lacZ* in *B. subtilis* (Fig. 5a) to be 1.3 ± 0.6 min and 2.6 ± 0.1 min, respectively, under a slow growth condition (Extended Data Fig. 9a,b), consistent with previous findings⁸. Despite this uncoupling, steady-state Z5 and Z3 levels were similar (Fig. 5b), implying negligible premature transcription termination. This finding is consistent with the dispensability of the Rho factor in

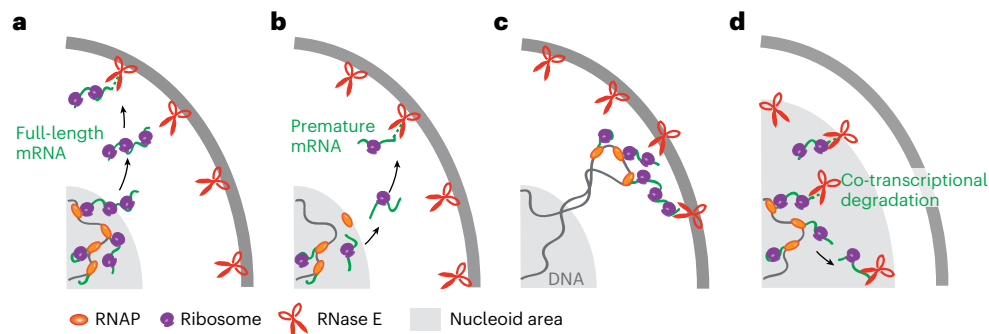


Fig. 6 | Generalizable model of mRNA degradation in bacteria. **a–c**, Proposed scenarios in *E. coli* (and possibly other bacterial species having the main ribonuclease on the membrane) for genes encoding cytoplasmic proteins with either a strong or a weak RBS (**a,b**) and for genes encoding inner-membrane

proteins (**c**). **d**, Proposed scenario in *C. crescentus* (and possibly other bacterial species having the main ribonuclease in the cytoplasm). The cartoon reflects that the nucleoid occupies a large area of the cytoplasm in *C. crescentus*⁷⁵.

B. subtilis, suggesting that Rho-dependent termination is probably rare in *B. subtilis*⁸.

Using rifampicin, we measured k_{d1} of $0.025 \pm 0.004 \text{ min}^{-1}$ and k_{d2} of $0.14 \pm 0.03 \text{ min}^{-1}$ for *lacZ* mRNA in *B. subtilis* (Fig. 5c and Extended Data Fig. 9c). Since premature transcription termination was not observed, k_{d1} probably reflects the true co-transcriptional degradation rate, yielding an estimated nascent mRNA lifetime of ~40 min ($1/k_{d1}$), much longer than the transcription time. Hence, as in *E. coli*, co-transcriptional degradation appears rare in *B. subtilis*, probably due to membrane localization of RNase Y, a homologue of *E. coli* RNase E⁴⁴.

Unlike *E. coli* and *B. subtilis*, *C. crescentus* and other α -proteobacteria have cytoplasmic RNase E^{13,14}. To investigate co-transcriptional degradation in *C. crescentus*, we used a strain with chromosomal *lacZ* under a xylose-inducible promoter⁴⁵ (Fig. 5d). The translation and transcription times were comparable ($2.5 \pm 0.2 \text{ min}$ versus $2.3 \pm 0.3 \text{ min}$; Fig. 5e,f and Extended Data Fig. 9d). When rifampicin was added at $t = 50 \text{ s}$, the Z5 signal decayed rapidly, reaching baseline before the transcription time (k_{d1} of $1.3 \pm 0.1 \text{ min}^{-1}$). Moreover, little 3' *lacZ* mRNA or LacZ protein accumulated (Fig. 5g and Extended Data Fig. 9e), indicating extensive premature transcription termination.

It is possible that the Rho factor is involved. Treatment with BCM⁴⁶ increased steady-state Z5 and Z3 levels (Extended Data Fig. 9f,g). Under BCM treatment, k_{d1} was lower than in untreated cells ($P = 1.8 \times 10^{-3}$; Extended Data Fig. 9h), probably due to the absence of fast-decaying prematurely released transcripts. Mathematical modelling estimated PT of 0.67, with k_{dPT} of 2.1 min^{-1} (Supplementary Discussion 3). Given the known interaction between the Rho factor and RNase E in *C. crescentus*⁴⁷, premature termination may be coupled with rapid mRNA turnover in this organism. We note that BCM only partially restored the Z3 signal (Extended Data Fig. 9h), suggesting that Rho-independent pathways and extensive co-transcriptional degradation may also contribute to reduced 3' mRNA levels.

Discussion

This study provides insights into the life cycle of bacterial mRNA by integrating transcription–translation coupling with spatial regulation of RNA decay. Our work highlights that RNAP–ribosome distance exists on a continuum across genes and species, with translation initiation rate emerging as a key determinant of gene-specific coupling in *E. coli*. Although transcription and translation completion times are similar, premature termination reveals that a substantial fraction of RNAPs can be uncoupled. Because Rho loading requires ~80 nt RNA⁴⁸, premature termination probably reflects a subset of uncoupled states exceeding this distance threshold. Structural and single-molecule studies have revealed that RNAP and the ribosome can physically couple over spacer lengths of ~10–100 nt^{49–52}, including configurations involving RNA

looping⁵³. Whether Rho can load onto looped RNA in these configurations and still cause premature termination remains an open question.

Given that RNAP–ribosome distance influences Rho loading, mechanisms that reduce this distance have been proposed. In *E. coli*, the canonical Shine–Dalgarno sequence (strong RBS) aligns closely with the consensus RNAP pause sequence³⁰, suggesting that strong RBS sequences may be accompanied by start-codon-proximal RNAP pausing, potentially reducing RNAP–ribosome distance early in transcription. However, genome-wide analysis of start-codon-proximal pause strength does not support early pausing as a global compensatory mechanism for weak translation initiation (Supplementary Discussion 4). Thus, while RNAP pausing may reinforce coupling under specific sequence contexts, translation initiation kinetics appear to be the dominant determinant governing coupling.

Comparison of *lacZ* RBS mutants with genome-wide translation initiation rates⁵⁴ suggests that at least 30% of endogenous genes fall within a slow translation initiation regime potentially associated with increased uncoupling (Extended Data Fig. 2b). Together with SEnd-seq analysis, this indicates that transcription–translation uncoupling is probably widespread in *E. coli*.

Spatial organization also shapes coordination between transcription and degradation. If the major ribonuclease is membrane localized, as in *E. coli* and *B. subtilis*, co-transcriptional degradation is probably negligible, and decay occurs primarily after transcript release (Fig. 6a,b). We found an exception to this rule for genes encoding inner-membrane proteins (Fig. 6c). Our data from *lacY2–lacZ* (with a strong RBS) showed that membrane proximity is necessary but not sufficient for co-transcriptional degradation of *lacY*. Additional factors probably promote RNase E action on *lacY* transcripts. For example, the signal recognition particle (SRP) and SecYEG, which mediate LacY translocation⁵⁵, might localize near RNase E and promote *lacY* mRNA degradation. This model suggests that membrane protein production is subject to tight spatial regulation, potentially enhancing cellular fitness given the limited membrane surface area relative to cytoplasmic volume and the high energetic cost of membrane channel proteins⁵⁶. In *C. crescentus*, which contains cytoplasmic RNase E, co-transcriptional degradation is prevalent⁵⁷, suggesting that subcellular localization of ribonucleases may influence the mRNA life cycle in a species-specific manner (Fig. 6d).

Our results further refine the role of translation in mRNA stability. Although ribosomes can directly protect mRNAs from RNase E^{6,32}, RNase E has been proposed to access cleavage sites by linear scanning along single-stranded RNA, and obstacles such as ribosomes or RNA base pairing can hinder this process³². In this framework, ribosome occupancy represents only one class of obstacle, which may explain why ribosome density alone is not a dominant determinant of mRNA stability. Transcripts with low ribosome density may gain protection if

increased ribosome spacing allows formation of RNA secondary structures. Indeed, a previous study reported an anticorrelation between TE and RNA structure⁵⁸. This tendency could counteract the expected effect of ribosome density on mRNA stability, potentially explaining the weak genome-wide correlation between TE and mRNA half-lives and the nearly constant k_{d2} values we observed across *lacZ* RBS mutants.

Crosstalk between transcription, translation and RNA decay can vary across biological systems. T7 transcription systems exhibit transcription–translation uncoupling due to the faster elongation rate of T7 RNAP⁵⁹ or the absence of native coupling interfaces with the host ribosome. However, T7 RNAP does not prematurely terminate⁶⁰. We therefore predict that transcripts made by T7 RNAP are degraded after transcription completion, as opposed to experiencing co-transcriptional degradation as previously proposed⁶¹.

In conclusion, our work identifies subcellular localization of RNase E (or its homologue) and RBS sequences as key determinants governing coordination between transcription, translation and mRNA degradation across genes and species. This framework provides a basis for quantitative modelling of gene expression across the genome⁵⁴ and for understanding the subcellular localization patterns of mRNAs across different genes and bacterial species^{13,62,63}. More broadly, these findings raise the possibility that spatial organization and translation kinetics may similarly shape coordination of transcription, translation and mRNA degradation in other systems lacking membrane-bound compartmentalization, such as archaea, chloroplasts and mitochondria.

Methods

Strains and plasmids

A complete list of strains and plasmids used in this study is provided in Supplementary Table 1, and oligonucleotide sequences and strain construction methods are provided in Supplementary Note. Plasmids were generated by Gibson ligation using PrimeSTAR HS DNA polymerase (Takara, R044A) for PCR and Gibson Assembly Master Mix (New England Biolabs (NEB), E2611). Chromosomal engineering was performed by lambda Red integration⁶⁴. Previously described strains and plasmids are referenced where appropriate in Supplementary Table 1. All engineered DNA regions were verified by sequencing. Oligonucleotides were obtained from Integrated DNA Technologies (IDT) unless otherwise noted. Strains and plasmids generated in this study are available from the corresponding author.

Strain growth conditions

For all gene expression measurements, cultures were prepared in two steps. First, colonies were inoculated in a liquid medium and grown for 8–9 h and diluted $>10^4$ -fold in fresh medium (same type as the initial culture) for overnight growth. Experiments were performed during exponential growth (optical density (OD)₆₀₀ = 0.2 for *E. coli* and *B. subtilis*; OD₆₆₀ = 0.2 for *C. crescentus*).

Unless otherwise noted, *E. coli* cells were grown in 20–30 ml M9 minimal medium (6 g l⁻¹ Na₂HPO₄, 3 g l⁻¹ KH₂PO₄, 0.5 g l⁻¹ NaCl, 1 g l⁻¹ NH₄Cl, 2 mM MgSO₄, 0.1 mM CaCl₂) supplemented with 0.2% glycerol, 0.1% casamino acids and 1 mg l⁻¹ thiamine at 30 °C using a water bath shaker. Under these conditions, most strains had doubling times similar to that of the wild type (95.0 ± 9.40 min), except RNase E ΔMTS (117.0 ± 27.0 min).

For the temperature-sensitive RNase E mutant (*rne3071*), cultures were grown as above until OD₆₀₀ = 0.2 and shifted to 43.5 °C for 10 min before induction⁶⁵.

B. subtilis cells were grown in MOPS minimal medium with maltose (1× MOPS mixture, 1.32 mM K₂HPO₄, 0.2% glutamate, 0.1 g l⁻¹ tryptophan, 0.4% maltose) at 30 °C (doubling time 92 ± 9.22 min).

C. crescentus cells were grown in M2G medium (0.87 g l⁻¹ Na₂HPO₄, 0.54 g l⁻¹ KH₂PO₄, 0.50 g l⁻¹ NH₄Cl, 0.5 mM MgSO₄, 0.5 mM CaCl₂, 0.01 mM FeSO₄, 0.2% glucose) at 28 °C (doubling time 142 ± 3.33 min).

Induction and re-repression of *lacZ* expression

For all *lacZ* expression under P_{lac} in *E. coli*, IPTG (0.2 mM) was added at $t = 0$. To re-repress *lacZ* expression, 500 mM glucose was added at $t = 75$ s (ref. 34). When bicyclomycin (BCM) was used, 100 µg ml⁻¹ BCM was added 5 min before IPTG addition.

The glucose addition time was chosen on the basis of two factors: First, it should be more than 1 min before T_3 , so that the time window i is sufficiently long for k_{di} fitting. Since we withdrew samples every 15–20 s, ~3–4 samples could be taken during the time window i of 1 min for k_{di} estimation. However, if glucose was added too early, not much Z5 signal developed. Considering these two factors, we decided to add glucose at $t = 75$ s for *lacZ* expression in our experimental condition. This glucose addition time was varied for a different gene (*araB*) and under different growth conditions because T_3 (transcription elongation time) changed. For example, glucose was added at $t = 50$ s for *rne3071* at 43.5 °C. For genes under P_{ara} (*lacZ* and *araB*), expression was induced with 0.2% arabinose at $t = 0$ and repressed with 500 mM glucose at $t = 75$ s for *lacZ* and $t = 60$ s for *araB*. Repression occurs through cAMP receptor protein-dependent regulation of the P_{ara} promoter.

For *lacZ* expression in *B. subtilis*, 5 mM IPTG was added at $t = 0$ and transcription initiation was inhibited with 200 µg ml⁻¹ rifampicin at $t = 30$ s. This rifampicin addition time was chosen on the basis of the same considerations used for glucose addition time.

For *lacZ* expression in *C. crescentus*, induction was performed with 0.3% D-xylose at $t = 0$, and transcription initiation was blocked with 200 µg ml⁻¹ rifampicin at $t = 50$ s. When BCM was used, 100 µg ml⁻¹ BCM was added 5 min before induction.

RNA extraction and RT–qPCR

At each time point, 0.4 ml of culture was withdrawn and immediately mixed with an equal volume of pre-cooled RNeasy Lysis Solution (Life Technologies). Samples were incubated on ice for 20 min and centrifuged. Pellets were resuspended in lysozyme solution (10 mg ml⁻¹ lysozyme in 10 mM Tris-HCl (pH 8) with 1 mM EDTA). Total RNA was extracted using the PureLink RNA mini kit (Life Technologies) with on-column DNase treatment. RNA was eluted in 50 µl RNase-free water. RNA concentrations were typically 60–80 ng µl⁻¹ (measured using a NanoDrop One Microvolume UV–Vis spectrophotometer). RT–qPCR was performed using KAPA SYBR FAST qPCR Master Mix (KAPA Biosystems) and a Bio-Rad CFX Connect system. Primer sequences are provided in Supplementary Table 2.

For very weak RBS strains (SK626 and SK637) and selected genes analysed in Fig. 4, Taqman probes were used. Complementary (c) DNA synthesis was performed using SuperScript III (Invitrogen), followed by RT–qPCR using TaqMan Universal Master Mix II with UNG (Applied Biosystems).

Analysis of time-course RT–qPCR data

Relative RNA levels were calculated as fold change relative to baseline (pre-induction samples) using the $2^{-\Delta C_t}$ method. Here, C_t denotes the cycle threshold at which the signal crosses the detection threshold.

T_5 and T_3 were obtained by linear fitting of the initial increase in the 5' and 3' signals and finding their intercept with the baseline. PT was calculated from the ratio of the initial 3' slope to the initial 5' slope under no-glucose experiments.

RNA degradation rates were obtained by fitting log-transformed RNA levels within defined time windows. For example, for data shown in Fig. 2c, Z5 values between $t = 150$ and 210 s were used to calculate k_{di} , and Z5 values between $t = 300$ and 600 s were used to calculate k_{d2} . Z3 values between $t = 300$ and 600 s could be used to calculate k_{d2} of Z3, but we focused our analysis on the degradation of Z5. These time windows were adjusted depending on transcription elongation time, which varies with gene length, RNAP speed and glucose addition time³⁴. Phenomenologically, t_5 , T_3 and t_3 were identified from the time when Z5 reached maximum, when Z3 started to appear from the baseline and

when Z3 reached maximum, respectively. All raw RT-qPCR data are provided in Source Data 1 and fitting results in Source Data 2.

Miller assay

LacZ activity was measured using a modified Miller assay³⁴. At each sampling time, 900 µl of cell culture was withdrawn and immediately mixed with a pre-cooled stop solution. The stop solution was 100 µl 5 mg ml⁻¹ chloramphenicol for *E. coli* and *C. crescentus*, and 100 µl 10 mg ml⁻¹ chloramphenicol and 10 mg ml⁻¹ erythromycin for *B. subtilis*. Cells were washed by centrifugation at 7,000 × *g* for 1 min (*E. coli*), 18,200 × *g* for 4 min (*B. subtilis*) or 6,000 × *g* for 4 min (*C. crescentus*) at 4 °C, and cell pellets were resuspended in 340 µl of 1× Z buffer (30 mM Na₂HPO₄, 10 mM NaH₂PO₄, 5 mM KCl, 0.5 mM MgSO₄) containing the same concentration of chloramphenicol (and erythromycin) as before the wash. Cells were lysed using SDS and β-mercaptoethanol.

For standard measurements, ortho-nitrophenyl-β-galactoside (ONPG) was used as the substrate, and absorbance was measured at OD₄₂₀ in a microplate reader (Synergy HTX multimode reader, BioTek). LacZ activity was calculated using standard Miller assay formulas based on OD₄₂₀, OD₅₅₀ and OD₆₀₀. For weak-RBS strains (SK421, SK626 and SK637) and non-*E. coli* species, the fluorogenic substrate MUG was used. In this case, LacZ activity was calculated from F350/460 and OD₆₀₀, measured from the plate reader.

Baseline activity was calculated from samples before LacZ induction and subtracted from all timepoints. Translation time (when the first LacZ proteins appeared) was determined by fitting the square root of LacZ activity to the initial linear increase⁶⁶.

When *lacZ* expression was re-repressed at *t* = 75 s, LacZ protein levels reached a plateau at *t* = 7 min. LacZ protein activity at the plateau was calculated by averaging data points between *t* = 7 and 10 min, and the difference from the baseline was used as a proxy for the total LacZ protein produced during the 75-s induction. When comparing LacZ protein levels across strains, we converted MUG-based LacZ measurements to the ONPG-based scale using the expression levels of SK98 and SK421 measured by MUG as an internal conversion control.

FISH microscopy

E. coli lacZ FISH was performed as described previously^{34,40}. For hybridization, we used 24 single-stranded DNA probes complementary to the first 1-kb region of *lacZ* mRNA (Z5_{FISH}). Each probe was labelled with a single Cy3B at the 5' end. Probe sequences are listed in Supplementary Table 3.

Phase-contrast and fluorescence images were taken in the Eclipse Ti-2 microscope (Nikon) equipped with Sola SE II 365 light engine (Lumencor), a phase-contrast objective Plan Apochromat (×100/1.45 NA, Nikon) and an Orca-R2 CCD camera (Hamamatsu Photonics). Cell outlines were detected using Oufiti⁶⁷, and fluorescent spots were identified using u-track⁶⁸ with Gaussian mixture-model fitting ($\alpha = 0.015$ for local maxima detection; $\alpha = 0.05$ for Gaussian peak testing). Custom MATLAB scripts were used to assign spots to individual cells and calculate spot number and localization (see Code availability). Spot positions were normalized to cell dimensions and visualized as 2D histograms. Number of cells and detected spots are summarized in Supplementary Table 4.

TE and PF analysis

Translation efficiency (TE) values were obtained from published ribosome profiling data (Supplementary Table 4 in ref. 25). Transcription progression factor (PF) values were calculated from published SEnd-seq data²⁸ (GEO access number GSE117737). For each protein-coding gene, upstream (from 0 to 200 nt downstream of the TSS) and downstream (from 500 nt downstream of the TSS to the end of the gene) zones were defined, and the ratio of downstream to upstream signal was used to calculate PF. If there was another qualified TSS located within the downstream zone, the region downstream of that TSS was excluded

from analysis. Genes with a short length (<600 bp) or a low expression level (<50 SEnd-seq reads) were removed from the analysis. TE and PF data are summarized in Source Data 3.

Rif-seq

Sample collection and RNA purification. *E. coli* wild-type cells (MG1655) were grown in MOPS EZ rich defined medium (Teknova, M2105) at 37 °C. An overnight culture was diluted into fresh medium to an initial OD₆₀₀ of 0.005 and grown to mid-log phase (OD₆₀₀ ≈ 0.3). Rifampicin (Sigma-Aldrich, R3501) was added at a final concentration of 500 µg ml⁻¹ to inhibit transcription initiation. Samples were collected at *t* = 0, 0.5, 1, 2, 3, 4, 5, 6, 8 and 10 min after rifampicin treatment by mixing 2 ml of culture with 250 µl of ice-cold stop solution (225 µl ethanol and 25 µl phenol). Cultures were centrifuged at 8,200 × *g* for 3 min, and pellets were flash frozen in liquid nitrogen.

Total RNA was purified from samples using hot acid phenol extraction. Briefly, frozen pellets were resuspended in 500 µl RNA extraction buffer (0.3 M NaOAc, 1 mM EDTA, 1% SDS) and mixed with 600 µl hot phenol (Invitrogen, AM9720) at 65 °C for 5 min. The mixture was centrifuged at 18,400 × *g* at 4 °C for 10 min, and the upper aqueous layer was transferred to a new tube. A second phenol/chloroform extraction was performed, followed by an additional extraction with an equal volume of chloroform. The upper layer was mixed with 2 µl GlycoBlue Coprecipitant (Invitrogen, AM9516) and an equal volume of isopropanol (Sigma-Aldrich, I9516) to precipitate RNA. RNA pellets were washed with 750 µl 80% ethanol, dried and resuspended in 10 mM Tris pH 7. Genomic DNA was removed using Turbo DNase (Invitrogen, AM2238).

RNA-seq library construction. Purified RNA was first fragmented by mixing with an equal volume of 2× alkaline fragmentation buffer (0.6 volumes of 100 mM Na₂CO₃ and 4.4 volumes of 100 mM NaHCO₃), incubated at 95 °C for 13 min and then purified using Zymo oligo CCkit (Zymo Research, D4060). Fragmented RNA (600 ng) was dephosphorylated using T4 polynucleotide kinase (NEB, M0201) at 37 °C for 1 h, followed by ligation to 3' linker oligos (sequences listed below) using T4 RNA ligase 2 truncated KQ (New England Biolabs, M0373) at 16 °C overnight.

3' linkers: /5rApp/NNNNNXXXXAGATCGGAAGAGCGTCGTGTAG GGAAAGA/3SpC3/

XXXX (4-base barcode): *t* = 0 CGAT; *t* = 0.5 GCTA; *t* = 1 ATCG; *t* = 2 TAGC; *t* = 3 GACT; *t* = 4 AGTC; *t* = 5 TCAG; *t* = 6 CTGA; *t* = 8 CATG; *t* = 10 TGAC.

The ligated products from different samples were pooled and purified by size excision on a Novex 10% TBE-urea polyacrylamide gel (Invitrogen, EC68752BOX). The purified pooled ligation products were then split into two tubes: one for total RNA and the other for mRNA enrichment. For samples requiring mRNA enrichment, ribosomal RNAs were removed from the total RNA with Ribo-Zero rRNA depletion kit (Epicentre, MRZGN126). For both total RNA-seq and mRNA-seq libraries, ligated RNA was reverse transcribed into cDNA by SuperScript IV Reverse Transcriptase (Invitrogen, 18090010) (reverse transcription primer 5'-TCTTTCCTACACGACGCTC-3'). The resulting cDNA was purified using Zymo Oligo CC kit. cDNA fragments were ligated to a second oligo (/5rApp/NNNNNCTGAAGATCGGAAGAGCACACGTCTGACTC/3ddC/) using T4 RNA ligase 1 (New England Biolabs, M0204) at 16 °C overnight. The final ligation products were purified by size excision on a Novex 10% TBE-urea polyacrylamide gel. Finally, the purified cDNAs were PCR amplified using Phusion polymerase (New England Biolabs, M0530) and the following PCR primers:

Forward primer:

5'-AATGATACGGCGACCGAGATCTACACTCTTCCCT ACACGACGCTC-3';

Reverse primer:

5'-CAAGCAGAAGACGGCATACGAGATCTTGTAGTGACTG-GAGTTCAGACGTGTGCTC-3' (of note, the 6-base index sequence CTTGTA can be replaced to allow multiplexing). After 8–10 cycles of

PCR amplification, libraries were size selected on a Novex 8% TBE gel (Invitrogen, EC6215BOX) and purified. Sequencing was performed on an Illumina HiSeq 4000 system, using single-end 50-nucleotide reads (Center for Advanced Technology, UCSF).

Sequencing data processing. The raw sequencing reads were first demultiplexed on the basis of the 4-base barcode corresponding to individual samples pooled in the library using `fastx_barcode_splitter` (https://github.com/agordon/fastx_toolkit). The linker sequence was clipped using `Cutadapt`⁶⁹. The clipped sequences were then mapped to the reference genome `NC_000913.3`, obtained from the NCBI Reference Sequence Database, using `STAR` (v.2.7.6a)⁷⁰. The following parameters were used:

```
-alignEndsType EndToEnd -outSAMmultNmax -1 -outFilter-MismatchNmax 2 -clip5pNbase 9 -outSAMtype BAM SortedByCoordinate -outWigType wiggle read1_5p -outWigStrand Stranded -outWigNorm None.
```

The `STAR` alignment output files are stranded wiggle files. ‘Signal.UniqueMultiple.str1’ and ‘Signal.UniqueMultiple.str2’ represent signal from both uniquely and multimapped reads mapped to the minus and plus strand, respectively. Multimapping signals were distributed equally across loci (for example, for ribosomal RNA reads).

Rif-seq data analysis. mRNA levels of individual genes before and after Rif treatment were calculated as follows. Since the overall levels of unstable RNAs decrease after Rif treatment due to degradation, we used a normalization standard to account for this change between timepoints. When *E. coli* is grown under optimal growth conditions, the majority of total RNA consists of ribosomal RNAs (rRNAs) and transfer RNAs (tRNAs), which are generally considered stable due to their significantly longer half-lives compared with mRNA. Using the total RNA-seq dataset, we divided all sequencing reads into two groups: stable RNAs, including rRNAs and tRNAs; and unstable RNAs, including mRNAs (mapped to protein-coding genes) and other non-coding RNAs (ncRNA). For each time point, we calculated the ratio of unstable RNA reads (mRNAs and ncRNAs) relative to the total stable RNA reads (rRNAs and tRNAs) as a normalization standard, referred to as ‘unstable RNA%’.

From total RNA-seq at each time point:

$$\text{unstable RNA\%} = \frac{\text{total number of unstable RNA reads}}{\text{total number of stable RNA reads}} \quad (1)$$

This unstable RNA% value was used to normalize the mRNA-seq reads per kilobase million (RPKM) values and to quantify the abundance of individual mRNAs across timepoints following Rif treatment. The adjusted mRNA RPKM (aRPKM) for gene *i* at a specific time point was calculated as:

From mRNA-seq at the same time point:

$$\text{aRPKM}_i = \frac{\text{number of reads mapped to gene } i \times 10^6}{\text{gene } i \text{ length (kb)} \times \text{total number of unstable RNA reads}} \times \text{unstable RNA\%} \quad (2)$$

Calculation of mRNA half-lives. We applied the following filters to select actively expressed protein-coding genes for half-life calculation: (1) the number of mRNA-seq reads mapped to the gene at time point $t = 0$ was at least 100 and (2) the gene had non-zero mRNA-seq read counts at all subsequent timepoints. A total of 996 protein-coding genes passed these filters.

If mRNAs decay exponentially, their half-lives can be calculated as follows:

$$R(t) = R_0 \times e^{-kt}; T_{1/2} = \ln(2)/k \quad (3)$$

where $R(t)$ is the RNA abundance at time t , R_0 is the RNA abundance at time 0 (before Rif treatment), k is the degradation rate constant, and $T_{1/2}$ is the RNA half-life.

Therefore, for each gene that passed the filters, we first normalized mRNA levels at different timepoints to the level at $t = 0$. We then applied a log transformation to these normalized values and fitted them to a linear model across the timepoints. To capture both early and late phases of mRNA decay, we performed piece-wise linear regression using the R package ‘segmented’, which can be found at <https://www.rdocumentation.org/packages/segmented/versions/2.1-2>. We used the ‘davies.test’ function to test for a change in slope. If the adjusted *P* value was lower than 0.05 (that is, two slopes are significantly different), the two separate half-lives along with the corresponding time breakpoint were calculated. Otherwise, the data were fitted into a single linear model, and the single half-life was calculated. The raw data and processed data are provided in Source Data 4.

RT-qPCR validation of Rif-seq decay profiles

To validate two-phase decay patterns observed in Rif-seq, decay kinetics were measured by RT-qPCR after rifampicin treatment. Cells (SK98 or SK626) were grown in MOPS EZ rich defined medium at 37 °C until $\text{OD}_{600} = 0.1$ and IPTG (0.2 mM) was added at $t = -20$ min. Rifampicin ($500 \mu\text{g ml}^{-1}$) was added at $t = 0$, and samples were collected over 10 min. When BCM was used, $100 \mu\text{g ml}^{-1}$ BCM was added at $t = -10$ min. RNA extraction and RT-qPCR were performed as described above using *Taqman* probes.

Simulation model for Rif-seq and two-phase decay pattern

To investigate the origin of the two-phase decay pattern observed in our Rif-seq data, we performed stochastic simulations of transcription and mRNA degradation using experimentally derived parameters. The simulations included RNAP initiation, elongation, premature termination and mRNA decay, and were carried out using an extended version of a previously developed totally asymmetric simple exclusion process (TASEP)-based framework⁷¹, in which premature termination was newly incorporated. In the model, we assumed that premature termination takes place in the middle of a 3,075-bp DNA template at a certain percentage (input parameter, PT). We had the first base of the mRNA (5′ end) decay either at k_{d1} (during transcription elongation, as nascent mRNA), at k_{dPT} (if prematurely released) or at k_{d2} (if transcription was completed at the gene end). This initial degradation was followed by 5′-to-3′ directional degradation. At 0.1-s intervals along the simulation, the copy number of 5′ mRNA (or any other positions) was counted. The mean copy number was calculated from 300 iterations of the simulation.

Similar to *lacZ* induction experiments, the simulated mRNA level increased from zero and reached a steady state before $t = 10$ min. Transcription initiation was halted (similar to Rif-seq) at $t = 20$ min of simulation time. For presentation, we plotted the log mRNA levels after transcription inhibition (re-defined this time as $t = 0$). Since the steady-state levels were different depending on the input parameters, log mRNA levels were shifted so that they start from 2 at $t = 0$ (Fig. 4d and Extended Data Fig. 8h,i).

Statistical test

All replicates represent biological replicates. Two-sample unpaired Student’s *t*-tests were performed using MATLAB (`ttest2` function). $P > 0.05$ is marked as NS for not significant. Statistical test results are summarized in Supplementary Table 5.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The Rif-seq data generated in this study have been deposited in GEO under accession number [GSE298132](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE298132). Processed Rif-seq and SEND-seq data as well as all RT-qPCR data are available in Source Data 1–4. Microscopy data reported in this paper can be found in Zenodo at <https://doi.org/10.5281/zenodo.17766371> (ref. 72). Primers and strain information can be found in the Supplementary Tables. Source data are provided with this paper.

Code availability

MATLAB code for data analysis and Python code for modelling are available in GitHub at https://github.com/sjkimlab/2026_NatMicro (ref. 73).

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Author contributions

Seunghyeon Kim, Y.Z. and Sangjin Kim conceived the idea. Seunghyeon Kim, Y.Z., X.J., A.H., Y.-H.W. and Sangjin Kim developed the methodology and carried out the investigation. All authors contributed to data interpretation and writing. Y.Z., S.L. and Sangjin Kim supervised the study and acquired funding.

Competing interests

The authors declare no competing interests.

Additional information

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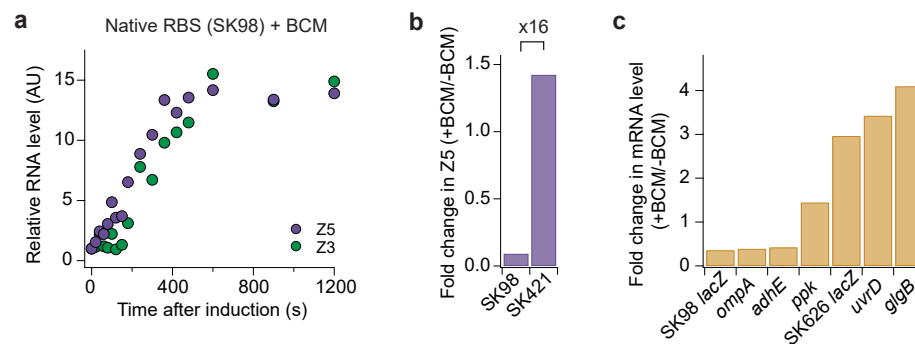
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Extended Data Table 1 | 5'UTR sequence of lacZ RBS mutants

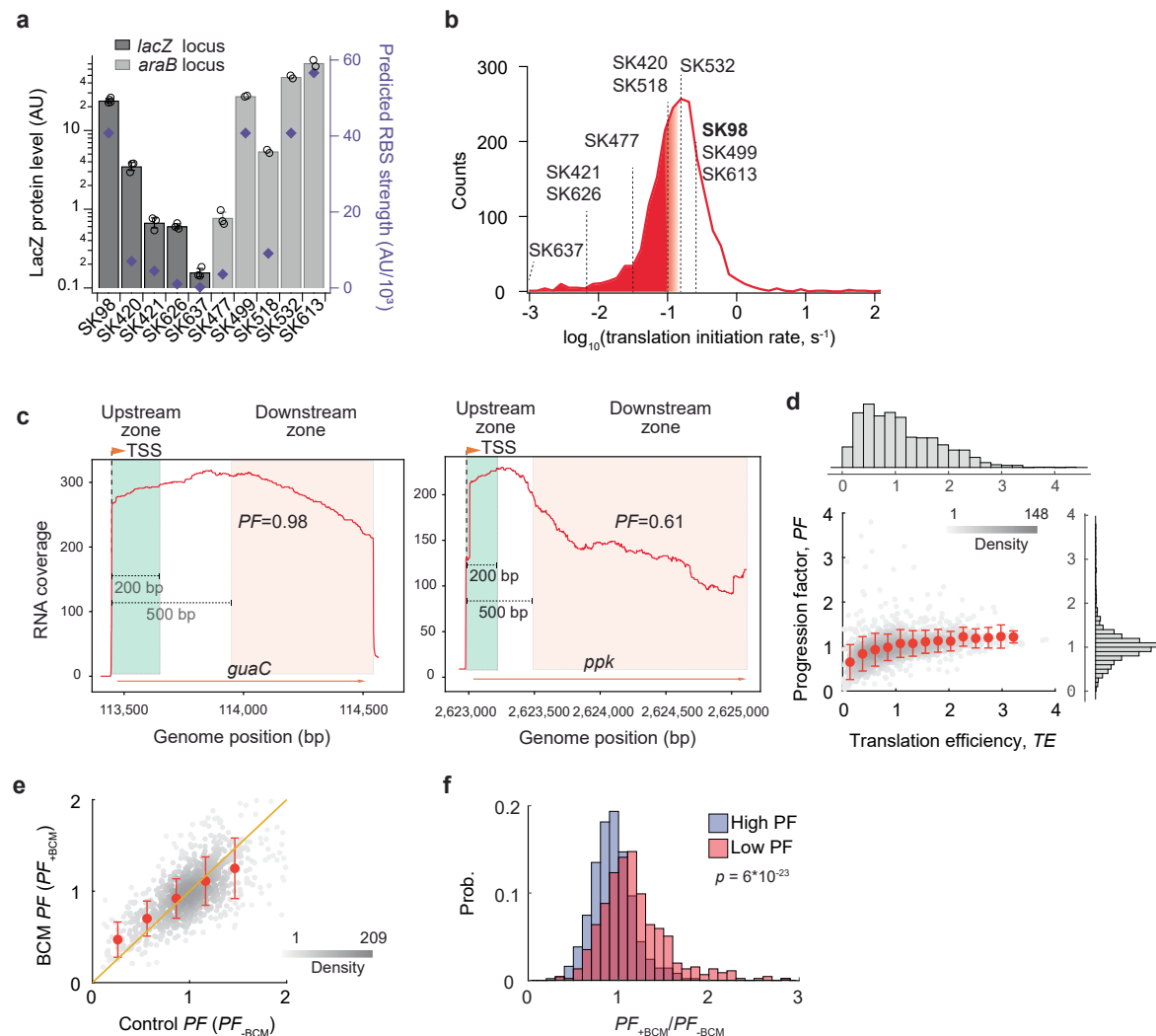
Strain	From +1 to start codon
SK98	attgtgagcggataacaatttcacacaggaaacagct ATG
SK420	attgtgagcggataacaatttcacactaagaccagct ATG
SK421	attgtgagcggataacaatttcacacggtgccagct ATG
SK477	acccgttttttggatggagtgaacg ATG
SK499	attgtgagcggataacaatttcacacaggaaacagct ATG
SK518	acccgttttttgaggagcactggacg ATG
SK532	attgtgagcggataacaatttcacacaggaaacagct ATG
SK613	acccgttttttgtaaggcaggatatt ATG
SK626	attgtgagcggataacaatttcacacggtgccagct GTG (accGTG)*
SK637	attgtgagcggataacaatttcacactaagaccagct GTG

*The original lacZ sequence has ATG as the start codon and ACC and ATG as the second and third codons, respectively. In SK626, both start codon and third codon were changed to GTG.



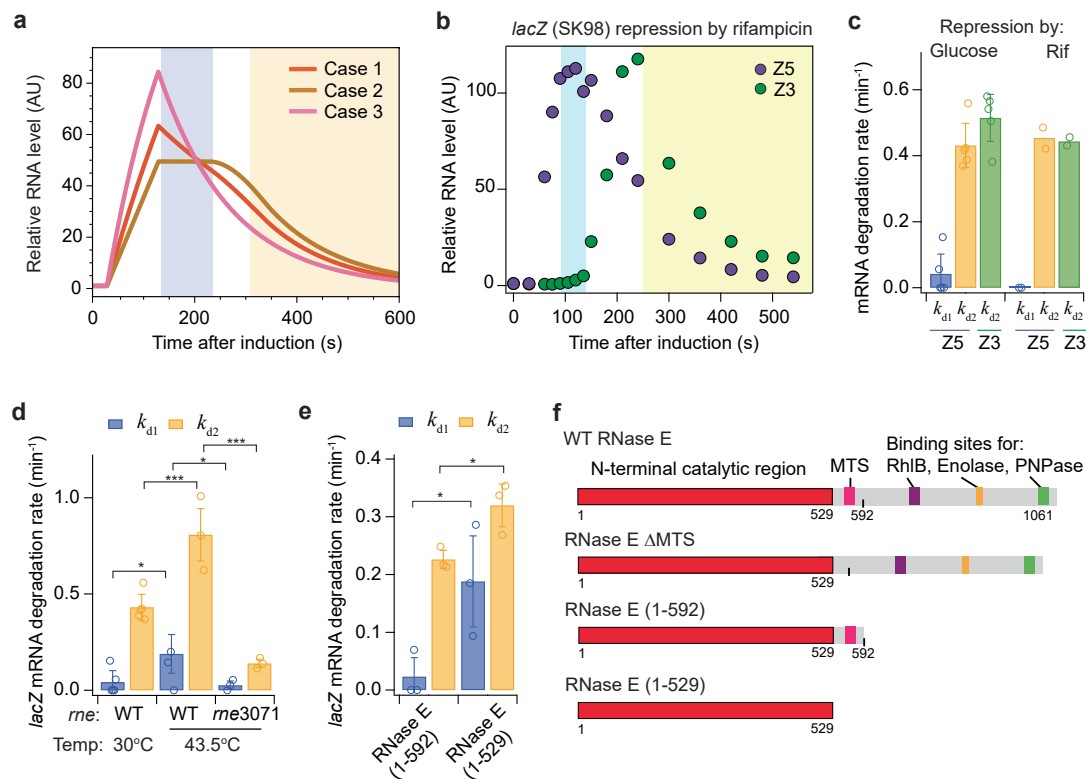
Extended Data Fig. 1 | Effect of BCM treatment on transcription levels. (a) Z5 and Z3 levels when *lacZ* transcription was induced at time zero in the native RBS strain (SK98) treated with BCM (100 $\mu\text{g}/\text{mL}$) 5 min before induction. Data are the mean of $n = 2$ biological replicates. (b) Fold difference in Z5 steady-state levels in BCM-treated and untreated cells from the experiments shown in (a). For steady-state levels, CT values measured at least 5 min after IPTG addition were used,

and fold change was calculated as $2^{\Delta\text{CT}}$, where $\Delta\text{CT} = \text{CT}_{\text{-BCM}} - \text{CT}_{\text{+BCM}}$. Steady-state levels were estimated from multiple time points collected from $n = 2$ biological replicates. Bar heights represent the ratio of the mean values. (c) Fold difference in mRNA steady-state levels in BCM-treated and untreated cells (data shown in Fig. 4e-h). Fold change was calculated as $2^{\Delta\text{CT}}$, using multiple steady-state samples collected from $n = 2$ biological replicates.



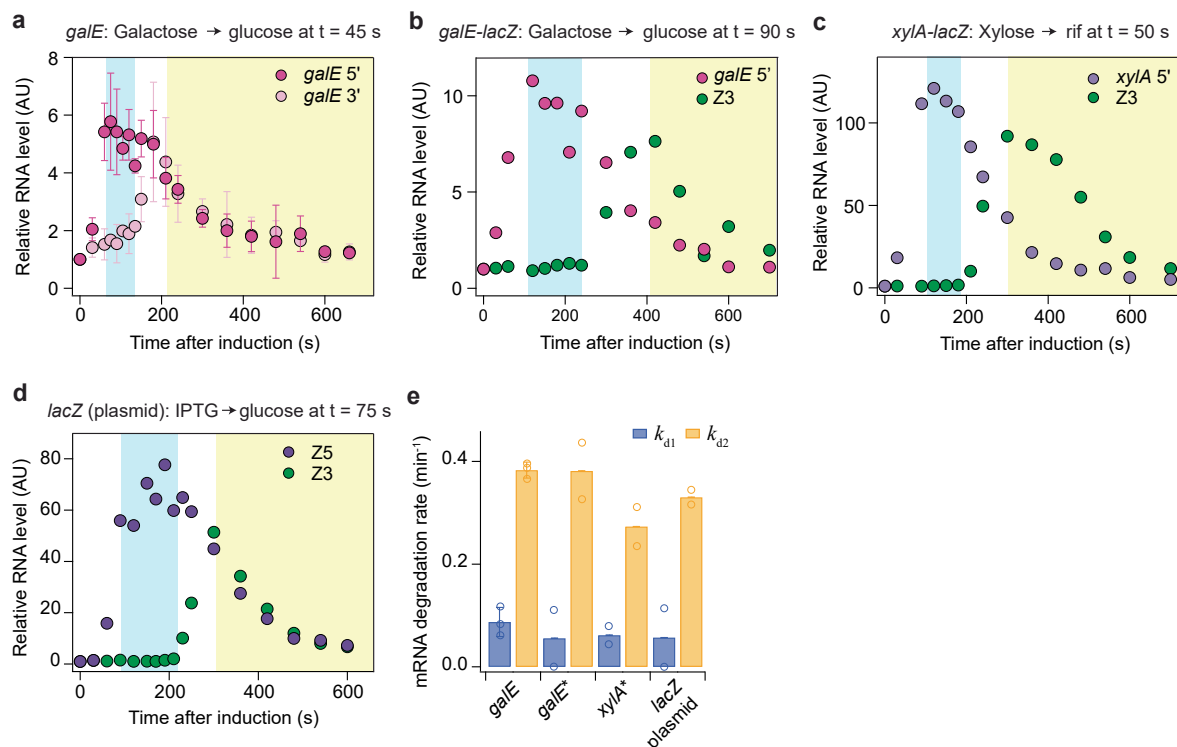
Extended Data Fig. 2 | Effect of RBS and translation activity on premature transcription termination. (a) LacZ protein production from various RBS mutant strains after 75-s induction, measured by the Miller assay. Data are mean $\pm$ s.d. from four (SK98) and three (SK420, SK421, SK626, SK637, SK477) biological replicates and the mean of biological duplicates (SK499, SK518, SK532, and SK613). RBS strength was calculated using the 25 bases upstream of the start codon and 35 bases of the coding sequence using an online tool²⁰. (b) Histogram of translation initiation rates under a slow-growth condition ($n = 1970$ genes; data from ref. 54). Translation initiation rates of our RBS mutant strains are marked relative to the original *lacZ* (SK98), whose translation initiation rate is 0.2 s^{-1} under a similar growth condition⁷⁶. The red shaded area indicates genes with low

translation initiation rates, for which we expect $PT > 0$ based on our RBS mutant strains. (c) Example SEnd-seq RNA coverage and the corresponding PF values. (d) Full-scale version of the density scatterplot shown in Fig. 1i. Red dots represent medians of binned data $\pm$ s.d. (e) Density dot plot of PF in control (-BCM) and BCM-treated (+BCM) *E. coli* cells for $n = 950$ genes. The unit of analysis is the gene. The yellow line indicates a 1:1 reference. Genes with low PF in the control condition ($PF < 1$) tend to show higher PF after BCM treatment. Red dots represent medians of binned data $\pm$ s.d. (f) Histograms of PF_{+BCM}/PF_{-BCM} (ratio) for low-PF genes ($PF_{-BCM} < 1$, $n = 454$) and high-PF genes ($PF_{-BCM} \geq 1$, $n = 496$). $p = 6 \times 10^{-23}$ was calculated using a two-sided two-sample Kolmogorov-Smirnov test.



Extended Data Fig. 3 | Co-transcriptional and post-transcriptional *lacZ* mRNA degradation. (a) Expected Z5 mRNA dynamics when *lacZ* expression is induced at $t = 0$ and re-repressed at $t = 75$ s. Case 1: distinct k_{d1} and k_{d2} ; Case 2: no co-transcriptional degradation, Case 3: $k_{d1} = k_{d2}$. See **Supplementary Discussion 2** for details. (b) Z5 and Z3 levels after induction at time zero and re-repression with rifampicin (500 μ g/mL) at $t = 50$ s. Data are the mean of biological duplicates. (c) *lacZ* mRNA degradation rates measured after repression with glucose or rifampicin. Data are mean $\pm$ s.d. from biological triplicates for glucose and the mean of biological duplicates for rifampicin. (d) *lacZ* mRNA degradation rates when RNase E is inactivated. The WT strain is SK98, and the *rne3071* strain is SK519. For repression, glucose was added at $t = 75$ s (30°C), $t = 30$ s (43.5°C for

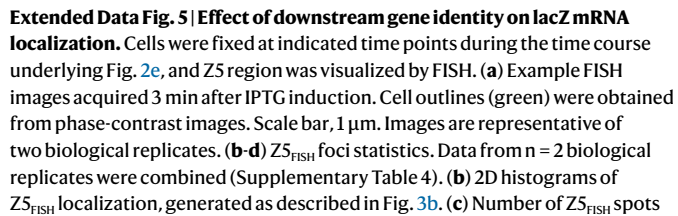
WT), or $t = 50$ s (43.5°C for *rne3071*). Data are presented as mean $\pm$ s.d. from $n = 5$ biological replicates at 30 °C and $n = 3$ biological replicates for 43.5 °C. k_{d1} , $*p = 0.018$ (SK98 at 30 °C and 43.5°C), $*p = 0.020$ (SK98 and SK519 at 43.5°C); k_{d2} , $***p = 9.9 \times 10^{-4}$ (SK98 at 30 °C and 43.5 °C), $***p = 4.2 \times 10^{-4}$ (SK98 and SK519 at 43.5 °C), from one-sided two-sample student t-tests. (e) k_{d1} and k_{d2} of *lacZ* mRNA in RNase E mutants localized either on the membrane or in the cytoplasm (strain SK370 and SK369). Data are presented as mean $\pm$ s.d. from $n = 3$ biological replicates. k_{d1} , $*p = 0.026$; k_{d2} , $*p = 0.015$ from one-sided two-sample student t-tests. (f) Linear representation of RNase E mutants used in Fig. 2d and in (e). Numbers indicate amino acid residues.



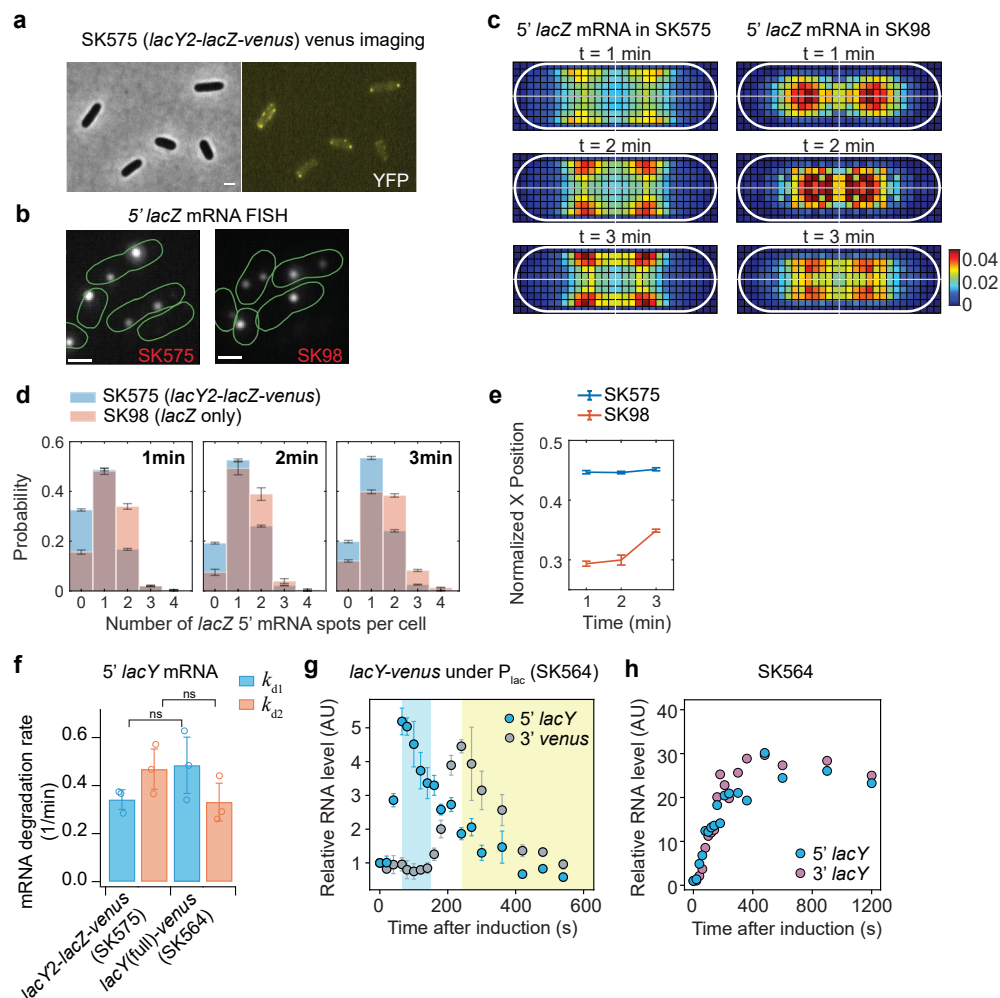
Extended Data Fig. 4 | Transient induction assay for genes other than *lacZ*.

(a) 5' and 3' *galE* mRNA levels after induction with 0.04% galactose at $t = 0$ and re-repression with 500 mM glucose at $t = 45$ s (SK98). RT-qPCR primers amplified the 40-145 nt and 838-973 nt regions of *galE* (gene length = 1017 nt). Data are mean $\pm$ s.d. from biological triplicates. (b) *galE-lacZ* fusion expressed from the native *galE* locus after induction with 0.04% galactose at $t = 0$ and re-repression with 500 mM glucose at $t = 90$ s (SK848; gene length = 4098 nt). Data are the mean of biological duplicates. (c) *xylA-lacZ* fusion expressed from the native *xylA* locus after induction with 0.04% xylose at $t = 0$ and re-repression with rifampicin

(500 $\mu\text{g}/\text{mL}$) at $t = 50$ s (SK850; gene length = 4404 nt). Primers amplified the 21-124 nt region of *xylA*. Data are the mean of biological duplicates. (d) Z5 and Z3 levels in SK356, carrying *lacZ* on a low-copy plasmid, after induction and re-repression as in Fig. 2c. Data are the mean of biological duplicates. (e) k_{d1} and k_{d2} values from (a-d). *galE** and *xylA** indicate endogenous genes fused to *lacZ*. Data are mean $\pm$ s.d. from biological triplicates for *galE* and the mean of biological duplicates for the others. Blue and yellow boxes indicate time windows i and iii used to calculate k_{d1} and k_{d2} , respectively.

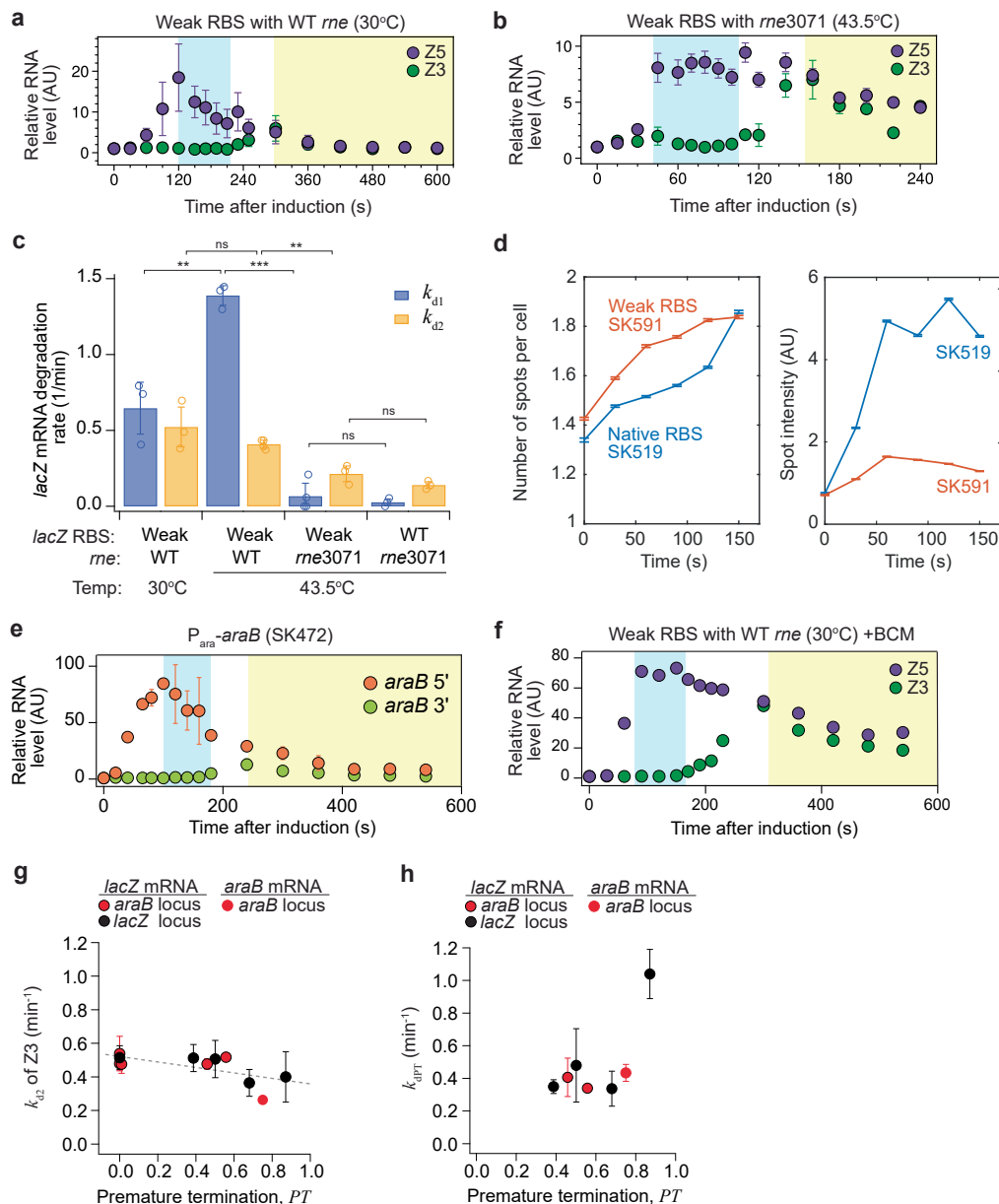


detected per cell. The number of mRNA spots was lower than the number of gene loci, consistent with ZS_{FISH} foci representing nascent transcripts at the DNA loci. The analysis included $n = 8885, 3629, 7056$ cells of SK390 at $t = 1, 2, 3$ min, and $n = 9653, 7213, 7090$ cells of SK435 at $t = 1, 2, 3$ min from two biological replicates. Data are presented as mean $\pm$ s.e. from bootstrapping. *lacZ* gene loci were imaged in SX259 grown under identical conditions at 30°C , and the mean was calculated from 12,265 cells from biological triplicates. **(d)** Mean position of ZS_{FISH} spots along the x-axis (short axis). Data represent the mean $\pm$ s.e. from bootstrapping.



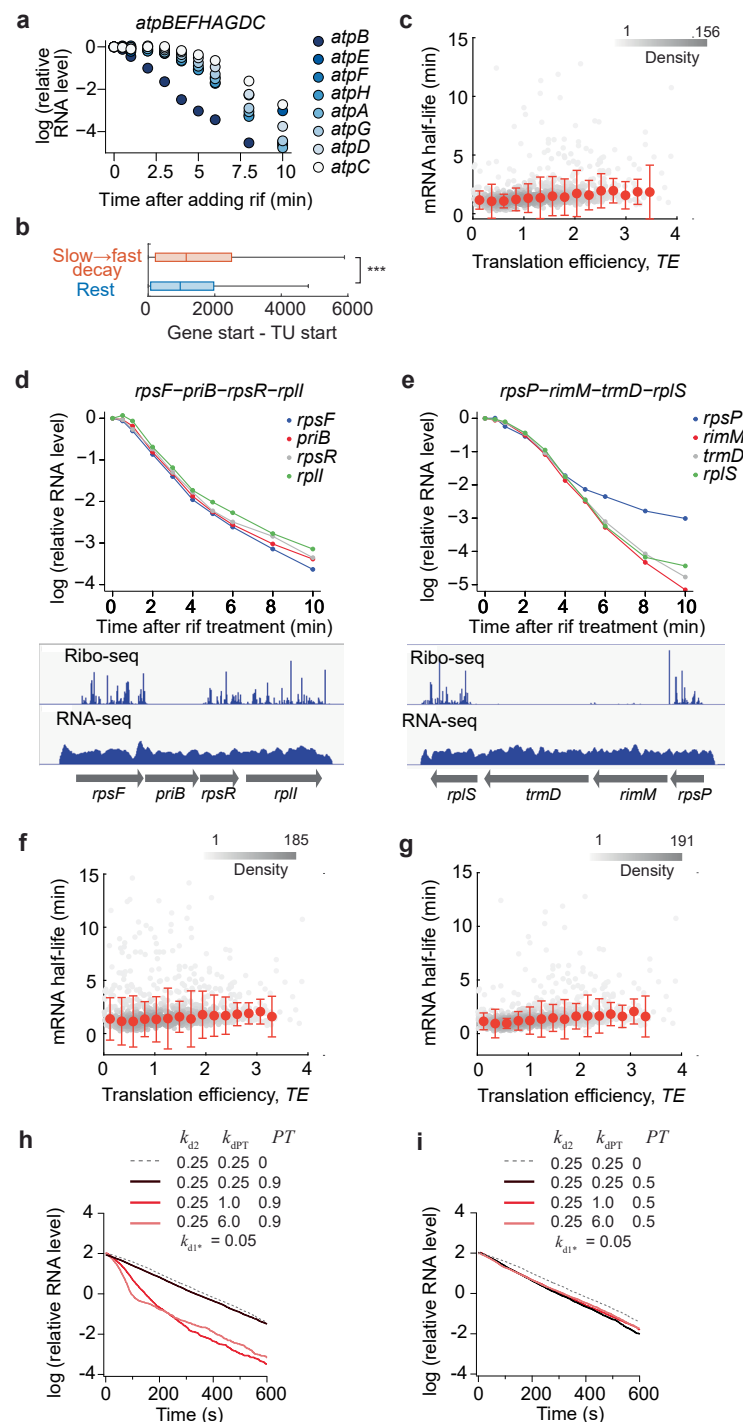
Extended Data Fig. 6 | Effect of transertion on mRNA localization and degradation. (a) Live-cell fluorescence image of LacY2-LacZ-Venus fusion protein in SK575 after 15-min induction. Venus signal localized to the cell periphery, supporting the translational fusion. Scale bar, 1 μ m. This experiment was performed once. (b) Example FISH images of 5' *lacZ* mRNA in SK575 and SK98 at t = 3 min after induction. Scale bar, 1 μ m. The images shown are representative of 4,000 cells from one biological replicate. (c-e) Z5_FISH results during the time course underlying Fig. 2f. Data from one biological replicate was used for the analysis (Supplementary Table 4). (c) 2D histogram of Z5_FISH localization. (d) Number of Z5_FISH spots detected per cell. The analysis included n = 6087, 9203, 5721 cells of SK575 at t = 1, 2, 3, min, and n = 1374, 370, 4047 cells of SK98 at

t = 1, 2, 3, min. Data are mean $\pm$ s.e. estimated by bootstrapping across cells. (e) Mean position of Z5_FISH spots along the x-axis (short axis). Data represent the mean $\pm$ s.e. from bootstrapping across cells. (f) k_{d1} and k_{d2} of 5' *lacY* region in SK575 and SK564. For repression, glucose was added at t = 75 s (SK575) or t = 50 s (SK564). Data are mean $\pm$ s.d. from biological triplicates. k_{d1} , $p = 0.18$; k_{d2} , $p = 0.17$ from two-sided two-sample t-tests. (g) Time-course 5' and 3' mRNA levels after 50-s pulse induction of *lacY-venus* under P_{lac} (SK564). Blue and yellow boxes indicate the time windows used to calculate k_{d1} and k_{d2} of 5' *lacY* mRNA. Data are mean $\pm$ s.d. from biological triplicates. (h) 5' and 3' *lacY* mRNA levels after induction. Similar steady-state levels of 5' and 3' *lacY* suggest negligible premature transcription termination. Data are the mean of biological duplicates.



Extended Data Fig. 7 | Effect of RBS strength on mRNA degradation. (a–b) Z5 and Z3 levels in weak-RBS *lacZ* strains: wild-type RNase E (SK421; a), temperature-sensitive RNase E (SK591; b). For repression, glucose was added at $t = 75$ s (a) and 50 s (b). Data are mean $\pm$ s.d. from $n = 3$ (a) and $n = 4$ (b) biological replicates. (c) Effect of RNase E on weak-RBS *lacZ* mRNA degradation. WT-RBS *rne3071* strain (SK519) is shown for comparison. Data are mean $\pm$ s.d. from $n = 3$ (SK421, 30°C), $n = 4$ (SK421, 43.5°C), $n = 4$ (SK591, 43.5°C), $n = 3$ (SK519, 43.5°C) biological replicates. For SK421 at 30°C versus 43.5°C, $**p = 0.023$ for k_{d1} , $p = 0.89$ for k_{d2} ; for SK421 versus SK591 at 43.5°C, $***p = 5.1 \times 10^{-6}$ for k_{d1} , $**p = 0.0014$ for k_{d2} ; for SK591 versus SK519 at 43.5°C, $p = 0.61$ for k_{d1} , $p = 0.068$ for k_{d2} , from one-sided two-sample t-tests. (d) Z5 FISH spot statistics during the time course described in Fig. 3b,c. Analysis included at least $n = 20,311$ cells (SK519) and $n = 24,363$ cells

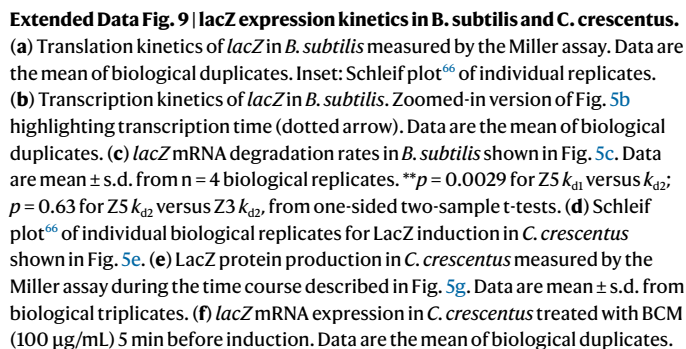
(SK591) from five biological replicates. Data are mean $\pm$ s.e. from bootstrapping. (e) *araB* mRNA levels during 50-s transient induction (SK472). Data are mean $\pm$ s.d. from biological triplicates. (f) Z5 and Z3 decay in BCM-treated weak RBS strain (SK421). Data are the mean of biological duplicates. (g) k_{d2} of 3' mRNA and *PT* for eleven strains. k_{d2} of 3' mRNA was calculated from Z3 decay in the time window iii. k_d data are mean $\pm$ s.d. from five (SK98) and three (all others) biological replicates; *PT* data are the mean of biological duplicates. Dashed line represents a linear fit. (h) k_{dPT} and *PT* of seven weak-RBS strains with detectable non-zero *PT*. k_d data are mean $\pm$ s.d. from biological triplicates; *PT* data are the mean of biological duplicates. In panel a, b, e, f, blue and yellow boxes indicate the time windows used to calculate k_{d1} and k_{d2} .



Extended Data Fig. 8 | Genome-wide investigation of mRNA half-lives in *E. coli* (Rif-seq).

(a) Rif-seq profiles for transcripts from the ATP synthase operon. **(b)** Distance between each gene start site and the transcription unit (TU) start site. Among 943 genes, $n = 294$ genes showing slow-to-fast decay were compared with the remaining genes ($n = 649$). Boxes indicate the interquartile range and median; whiskers indicate the minimum and maximum values. *** $p = 2.9 \times 10^{-4}$ from a one-sided two-sample t-test. **(c)** Full-scale version of the density scatterplot shown in Fig. 4b with $n = 820$ genes. Red dots represent median $\pm$ s.d. of binned data. **(d-e)** Example Rif-seq and ribosome profiling (Ribo-seq) profiles in operons. Genes within the same operon can exhibit different ribosome densities but show either similar (d) or different (e) mRNA decay rates. **(f-g)** Density scatterplots of translation efficiency (*TE*) and mRNA half-life

for all genes ($n = 943$). For genes with two-phase decay, either the late-phase (slow) half-life (f) or the early-phase (fast) half-life (g) is shown. Red dots represent median $\pm$ s.d. of binned data. **(h-i)** Simulated 5' mRNA decay as in Rif-seq. A two-phase decay pattern is apparent when $k_{dPT} > k_{d2}$ (h) and $PT > 0.5$ (i). If one of these conditions is not satisfied, the decay appears as one-phase. For example, when k_{dPT} is close to k_{d2} , the decay of prematurely terminated RNA and that of fully transcribed RNA are not distinguishable (that is, the total RNA decays at one decay rate). Also, when $k_{dPT} > k_{d2}$ but PT is low, the fraction of prematurely terminated RNA is relatively low, so that its distinct (fast) decay is not apparent in total RNA decay. Altogether, genes with two-phase decay are likely a subset of genes experiencing pronounced PT and $k_{dPT} > k_{d2}$.



Nature Microbiology

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| ☐ | ☒ | The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
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<i>Give P values as exact values whenever suitable.</i> |
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Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	A commercial software (Nikon NIS-Elements version 5.11.03) was used for image acquisition.
Data analysis	Open-source software packages Oufiti and u-track (version 2.3) were used for FISH image analysis. For piece-wise linear regression, the R package segmented (open-source) was used. Microsoft Excel for Microsoft 365 MSO and Igor Pro 9.04 were used for qRT PCR data analysis. Other home-built code is available from github.com/sjkimlab/2026_NatMicro .

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All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

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The Rif-seq data generated in this study have been deposited in GEO under accession number GSE298132. Processed Rif-seq and SEnd-seq data as well as all qRT-

PCR data are available in Source Data 1-4. Microscopy data reported in this paper can be found in a public repository: <https://doi.org/10.5281/zenodo.17766371>. Primers and strain information can be found in the Supplementary Tables.

Research involving human participants, their data, or biological material

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Reporting on sex and gender	N/A
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	As customary in the field, we did not perform statistical analysis to predetermine sample size. For RT-qPCR experiments, 2-5 independent biological replicates were used for each condition. For smFISH experiments, 1-5 independent biological replicates were used, with total conducted, with 300-30,000 individual cells included in the analysis. These sample sizes ensure statistical robustness and are consistent with common practice in the field.
Data exclusions	In RNA sequencing data analysis, genes failing sequencing quality control due to low read counts were excluded. No other data were excluded.
Replication	We performed multiple replicates as stated in the Sample Size section. Measurements were reliably reproduced with 2-5 replicates.
Randomization	No formal randomization was used. For qRT-PCR, biological replicate cultures were grown and processed independently on different days to reduce batch effects. For smFISH, imaging fields were selected using a raster-scanning pattern to minimize selection bias. For certain RNA-seq analyses, genes were grouped according to predefined criteria described in the main text and figure legends.
Blinding	Blinding was not applicable to this study, as it did not involve clinical data or human or animal models. The experiments relied on automated analysis pipelines, which minimize human bias. The quantitative measurements performed in this study can be independently verified through repeated experiments, further reducing the potential for observer bias.

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☒	☐ Eukaryotic cell lines
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☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
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Methods

n/a	Involvement in the study
☒	☐ ChIP-seq
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Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A